

EDUCATIVE COMMENTARY ON ISI 2024 MATHEMATICS PAPERS

(Uploaded on 12/07/2024)

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Introduction

The original motivation for writing the annual commentaries on ISI B.Stat. Entrance Tests is now waning. They began in 2016 because it was the year in which some exciting questions came to the notice of the author. Later on commentaries from 2014 were added. Unfortunately, after 2017, there was a noticeable decline in the quality of questions. Some readers also brought to the notice of the author that a few questions which were hailed in the commentary were picked from some reputed international contests. This did not detract from the intrinsic quality of the questions. Nor is it likely that some candidates had seen the solutions already. But apart from loss of originality, it is an insult to a good problem (for which originally ample time was allowed) to expect the candidates to crack it in a couple of minutes.

Still, the annual commentaries on ISI Entrance Test continued more as a mission, because some students pointed out that unlike the JEE, there are no sources to get the answers of ISI Entrance tests. But things are changing here too. In addition to the official key to the multiple choice questions in Paper 1, videos giving worked out solutions are available. As for the second paper, solutions for some of the problems are presented on the Artofsolvingproblems.

But even with both the justifications for the commentaries losing their strengths, I believe there is still some room to bring out the educative aspects of some of the problems and solutions. So, here is the commentary on ISI 2024.

I thank several contributors to the solution to Q.10 of Paper 1. Thanks

are also due to Siddhesh Naik for helping me out with Q.27 in Paper 1. But, above all, I thank the readers of the previous commentaries who eagerly awaited the present commentary. Their encouraging feedback keeps me going.

As in the other commentaries, unless otherwise stated, all the references are to the author's *Educative JEE Mathematics*, Third Edition, published by Universities Press, Hyderabad.

Readers are invited to send their comments and point out errors, if any, in the commentary. They may be sent either by e-mail (kdjoshi314@gmail.com) or by an SMS or a WhatsApp message on mobile (9819961036).

ISI BStat-BMath-UGA-2024 Paper 1 with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

Q.1 Let j be a number selected at random from $\{1, 2, \dots, 2024\}$. What is the probability that j is divisible by 9 and 15?

- (A) $\frac{1}{23}$ (B) $\frac{1}{46}$ (C) $\frac{1}{44}$ (D) $\frac{1}{253}$

Answer and Comments: (B). Divisibility by both 9 and 15 is equivalent to divisibility by their l.c.m. (the least common multiple) which is 45. So the question amounts to finding the number of multiples of 45 in the given set. Obviously, this is the integral part of $\frac{2024}{45}$. By a simple division, $2024 = 44 \times 45 + 44$. Hence there are 44 integers in the given set which are divisible by both 9 and 15. Therefore the desired probability is $\frac{44}{2024} = \frac{1}{45 + 1} + \frac{1}{46}$.

An absolutely simple problem once the idea of l.c.m. strikes.

Q.2 Let S_n be the set of all n -digit numbers whose digits are all 1 and 2 and there are no consecutive 2's. (Example: 112 is in S_3 but 221 is not in S_3 .) Then the number of elements in S_{10} is

- (A) 512 (B) 256 (C) 144 (D) 89

Answer and Comments: (C). Let a_n denote the number of elements in S_n . It would be great if we can express a_n by an explicit formula involving n (e.g. $n^2 - 3n$ or 2^{n-1} or $n!$) because in that case all we need to do is to put $n = 10$ in that formula. But things are often not so simple. The next best thing is to find a_n by hand for some small values of n , guess some pattern and prove the guess by induction on n . Thus, for $n = 1, 2, 3, 4$ we have

$$\begin{aligned} S_1 &= \{1, 2\} \\ S_2 &= \{11, 12, 21\} \\ S_3 &= \{111, 112, 121, 211, 212\} \\ \text{and } S_4 &= \{1111, 1112, 1121, 1211, 1212, 2111, 2112, 2121\} \end{aligned}$$

which shows

$$a_1 = 2, \ a_2 = 3, \ a_3 = 5, \ \text{and} \ a_4 = 8 \quad (1)$$

If we like, we can add a dummy initial term $a_0 = 1$ (which is actually consistent because S_0 is *not* the empty set. It consists of the null sequence and hence has one element in it.)

Whether we include the dummy term or not, there is a clear pattern in (1). Every term, except the first two is a sum of its two preceding terms. (The advantage of the dummy term a_0 is that the pattern is consistent with it and enhances the hope that it will continue for larger values of n too. If we drop a_0 , we shall have to calculate a_5 to strengthen our guess.)

If we assume for a moment that this relationship, viz.

$$a_n = a_{n-1} + a_{n-2} \quad (2)$$

is valid for all $n \geq 3$, then we can calculate a_n , one by one, till we reach a_{10} . That is,

$$a_5 = a_4 + a_3 = 8 + 5 = 13 \quad (3)$$

$$a_6 = 13 + 8 = 21 \quad (4)$$

$$a_7 = 21 + 13 = 34 \quad (5)$$

$$a_8 = 34 + 21 = 55 \quad (6)$$

$$a_9 = 55 + 34 = 89 \quad (7)$$

$$\text{and finally, } a_{10} = 89 + 55 = 144 \quad (8)$$

In a multiple choice question, it is a safe bet to mark (C) as the correct answer. For an honest answer, we must prove (2). This can be done by classifying elements of S_n into two types. Note that every number in S_n must end either with 1 or with 12. A number of the former type can be obtained by adding a 1 at the end of any number in S_{n-1} . Similarly, any number of the latter type can be obtained by adding 12 at the end of any number in S_{n-2} . As these operations are bijective, we get (2).

Even if we have proved (2), it is hardly practical to use it if we wanted a_{100} instead of a_{10} . For that we need to express a_n solely as a function

of n . This is possible and the answer is

$$a_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^{n+2} - \left(\frac{1 - \sqrt{5}}{2} \right)^{n+2} \right] \quad (9)$$

Using (2) and induction on n , it can be *verified* that this answer is correct. But *arriving at* it is considerably more difficult. (2) is popularly known as **Fibonacci relation**. See Comment No. 11 of Chapter 4 for more details.

Q.3 Let T be a right angled triangle in the plane whose side lengths are in geometric progression. Let $n(T)$ denote the number of sides of T that have integer lengths. Then the maximum value of $n(T)$ over all such T is

(A) 0 (B) 1 (C) 2 (D) 3

Answer and Comments: (B). The diction of the problem is somewhat clumsy. A better wording would have been to ask for the maximum possible number of integral sides in a right angled triangle, given that the lengths of the sides are in a geometric progression. If this last restriction were not there then the answer is 3 because there are infinitely many right angled triangles with all three sides having integer lengths. (See Comment No. 3 of Chapter 8.)

Because of this restriction, we take the three sides as a, ar and ar^2 with the last representing the length of the hypotenuse. Then the Pythagoras theorem gives

$$a^2 + a^2 r^2 = a^2 r^4 \quad (1)$$

This reduces to a quadratic in r^2 , solving which we get

$$r^2 = \frac{1 \pm \sqrt{5}}{2} \quad (2)$$

As $r^2 > 0$, we discard the negative square root and get

$$r = \sqrt{\frac{1 + \sqrt{5}}{2}} \quad (3)$$

Note that not only r but r^2 is also irrational. Hence out of the three sides of lengths a, ar and ar^2 , at most one can be an integer. But we are free to let any one side to be an integer and then determine the other two using (2). For example, we take $a = 1, b = r$ and $c = r^2$ with r as above. Hence there is a triangle satisfying the restrictions with one integer side, but no such triangle with two (or three) integral sides.

An unusual combination of geometry and quadratic equations.

Q.4 If $x = 1 + \sqrt[5]{2} + \sqrt[5]{4} + \sqrt[5]{8} + \sqrt[5]{16}$, then the value of $(1 + \frac{1}{x})^{30}$ is

(A) 2 (B) 5 (C) 32 (D) 64

Answer and Comments: (D). Put $u = \sqrt[5]{2} = 2^{1/5}$. Then $u^5 = 2$. And

$$\begin{aligned} x &= 1 + u + u^2 + u^3 + u^4 \\ &= \frac{1 - u^5}{1 - u} \\ &= \frac{1}{u - 1} \end{aligned} \tag{1}$$

Hence,

$$1 + \frac{1}{x} = 1 + u - 1 = u \tag{2}$$

Therefore

$$(1 + \frac{1}{x})^{30} = u^{30} = (u^5)^6 = 2^6 = 64 \tag{3}$$

A simple problem once the correct substitution strikes.

Q.5 There are 30 True or False questions in an examination. A student knows the answer to 20 questions and guesses the answers to the remaining 10 questions at random. What is the probability that the student gets exactly 24 correct answers.

(A) $\frac{105}{2^9}$ (B) $\frac{105}{2^8}$ (C) $\frac{105}{2^{10}}$ (D) $\frac{4}{2^{10}}$

Answer and Comments: (A). We might as well forget the 20 questions whose answers the student knows as this, in no way, affects the desired probability. So the problem is essentially to find the probability of exactly 4 correct answers out of 10 answers given at random. This is a simple binomial probability. The four questions can be chosen from ten questions (whose answers the student does not know) in $\binom{10}{4} = \frac{10 \times 9 \times 8 \times 7}{4!} = 210$ ways. For each such choice, the probability that the chosen questions are correctly answered while the others are wrongly answered is $\frac{1}{2^{10}}$. Hence the desired probability is $210 \times \frac{1}{2^{10}} = \frac{105}{2^9}$.

(In absence of any other information, we assume that the probability that a guess is correct is $\frac{1}{2}$. If, instead, this probability were some number p between 0 and 1, then the answer would be $\frac{210}{p^4(1-p)^6}$.)

Q.6 If the points z_1 and z_2 are on the circles $|z| = 2$ and $|z| = 3$ respectively, and the angle included between these vectors is 60° , then the value of $\frac{|z_1+z_2|}{|z_1-z_2|}$ is

(A) $\sqrt{\frac{19}{7}}$ (B) $\sqrt{19}$ (C) $\sqrt{7}$ (D) $\sqrt{\frac{7}{19}}$

Answer and Comments: (A).

z_1 and z_2 will be of the form $2e^{i\theta}$ and $3e^{i(\theta+\frac{\pi}{3})}$ respectively for some θ (specifically, the argument of z_1). As we are interested only in the ratio of the absolute values, we may suppose, without loss of generality, that $\theta = 0$. Then

$$\begin{aligned} z_1 &= 2 \\ \text{and } z_2 &= 3e^{i\pi/3} \\ &= 3\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) \\ &= \frac{3}{2} + i\frac{3\sqrt{3}}{2} \end{aligned} \tag{1}$$

By a direct calculation,

$$|z_1 + z_2| = \left| \frac{7}{2} + i \frac{3\sqrt{3}}{2} \right| = \frac{1}{2} \sqrt{49 + 27} = \sqrt{19} \quad (2)$$

Similarly,

$$|z_1 - z_2| = \left| \frac{1}{2} - i \frac{3\sqrt{3}}{2} \right| = \frac{1}{2} \sqrt{1 + 27} = \sqrt{7} \quad (3)$$

The answer is obtained by dividing (2) by (3).

A simple problem on complex numbers. Although a complex number can be regarded as a vector (specifically, the position vector of the point in the complex plane represented by it), doing so is not such a common practice. (Possibly because it is a sort of an algebraic degradation. There are many things you can do with complex numbers, such as their multiplication, which have no analogue for vectors.) Candidates not familiar with this practice are likely to be confused by the words, ‘these vectors’ and may have to spend some time to figure out exactly what is meant.

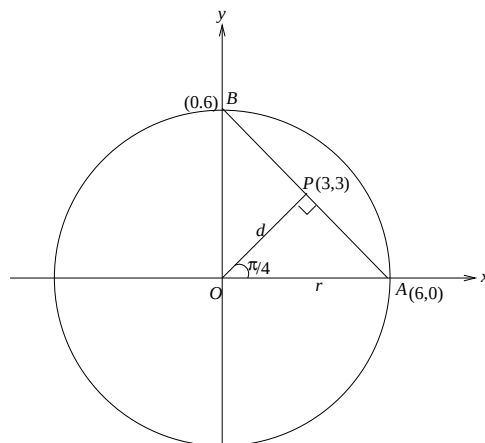
Q.7 The angle subtended at the origin by the common chord of the circles $x^2 + y^2 - 6x - 6y = 0$ and $x^2 + y^2 = 36$ is

- (A) $\pi/2$ (B) $\pi/4$ (C) $\pi/3$ (D) $2\pi/3$

Answer and Comments: (A). Note that the origin is also the centre of the second circle. There are readymade but rather complicated formulas for the angle a chord of a circle subtends at its centre. Instead we note that it suffices to find the perpendicular distance, say d , of the chord from the centre because the radius r of the second circle is already known to be 6.

The equation of the common chord, say AB , is obtained by subtracting the two equations. It comes out as

$$x + y = 6 \quad (1)$$



Hence d , the perpendicular distance of O from the chord is

$$d = \frac{6}{\sqrt{1+1}} = \frac{6}{\sqrt{2}} = 3\sqrt{2} \quad (2)$$

(Alternately, let P be the foot of the perpendicular from O to the chord. As the equations of the circle as well as of the chord are symmetric in x and y , P must be $(3, 3)$ and hence $d = OP = \sqrt{9+9} = 3\sqrt{2}$.)

Hence the angle between the perpendicular OP and the radius OA (or OB) is $\cos^{-1}(\frac{3\sqrt{2}}{6}) = \cos^{-1}(\frac{1}{\sqrt{2}}) = \pi/4$. The angle which AB subtends at O is twice this.

The first circle has little role in the problem except to identify a particular chord of the second circle. Once the equation of this chord is determined as $x + y = 6$, even by inspection, we can identify $(6, 0)$ and $(0, 6)$ as the end points of the chord. In that case, the angle subtended at O is 90 degrees without any further work. The numerical data is probably designed to reward those who can see this short cut.

Q.8 The precise interval on which the function $f(x) = \log_{1/2}(x^2 - 2x - 3)$ is monotonically increasing is

- | | |
|---------------------|--------------------|
| (A) $(-\infty, -1)$ | (B) $(-\infty, 1)$ |
| (C) $(1, \infty)$ | (D) $(3, \infty)$ |

Answer and Comments: (A). The natural logarithm function is monotonically increasing. As a result, if a (positive valued) function

$f(x)$ is increasing/decreasing on some interval (a, b) then $\ln f(x)$ also behaves the same way on (a, b) . But when the base of the logarithm is less than 1, as in the present problem, the things are reversed. That is, $\log_{1/2} f(x)$ is increasing/decreasing on (a, b) according as $f(x)$ is decreasing/increasing on it. So the present problem boils down to finding the interval (a, b) on which the function $f(x) = x^2 - 2x - 3$ is (i) positive valued and (ii) monotonically decreasing. The first part follows by factoring $f(x)$ as $(x - 3)(x + 1)$. It is positive valued for $x > 3$ and for $x < -1$. However, on $(-\infty, -1)$, it is monotonically decreasing while on $(3, \infty)$ it is monotonically increasing. So the precise interval which satisfies both (i) and (ii) is $(-\infty, -1)$.

Q.9 Let $x_1, x_2, \dots, x_n$ be non-negative real numbers such that $\sum_{i=1}^n x_i = 1$.

What is the maximum value of $\sum_{i=1}^n \sqrt{x_i}$?

- (A) 1 (B) $\sqrt{n}$ (C) $n^{3/4}$ (D) n

Answer and Comments: (B). The function $f(x) = \sqrt{x}$ is concave downwards on $(0, \infty)$ as can be seen from its second derivative. So, for any non-negative real numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ which add to 1, we have, by Jensen's inequality (Comment No. 18 of Chapter 13),

$$f(\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n) \geq \alpha_1 f(x_1) + \alpha_2 f(x_2) + \dots + \alpha_n f(x_n) \quad (1)$$

We take each α_i as $\frac{1}{n}$. Then (1) reduces to

$$\sqrt{\frac{x_1 + x_2 + \dots + x_n}{n}} \geq \frac{\sqrt{x_1} + \sqrt{x_2} + \dots + \sqrt{x_n}}{n} \quad (2)$$

Using $x_1 + x_2 + \dots + x_n = 1$ and simplifying, we get

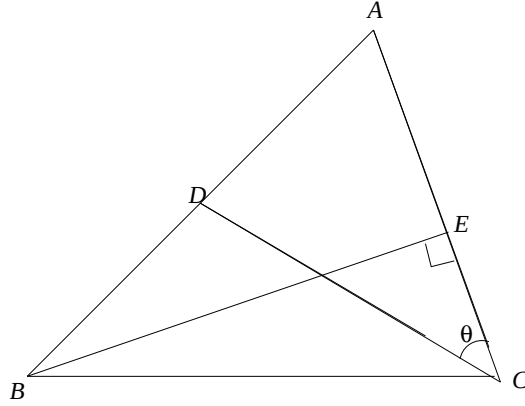
$$\sqrt{n} \geq \sqrt{x_1} + \sqrt{x_2} + \dots + \sqrt{x_n} \quad (3)$$

Since equality holds when each $x_i = \frac{1}{n}$, we see that $\sqrt{n}$ is the maximum possible value of $\sum_{i=1}^n \sqrt{x_i}$ under the given restrictions on x_i 's.

The result can also be derived using other inequalities such as the Cauchy-Schwarz inequality or the root mean square inequality. But

most of these inequalities can be derived from Jensen's inequalities. A far more advanced technique is to use Lagrange's multipliers (see the end of Comment No. 14, Chapter 13.) We have preferred the Jensen's inequality as the most general within the scope of JEE.

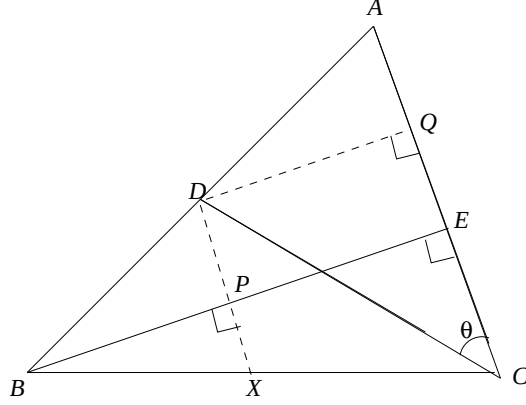
- Q.10 In $\triangle ABC$, CD is the median and BE is the altitude. Given that $\overline{CD} = \overline{BE}$, what is the value of $\angle ACD$?



- (A) $\pi/3$ (B) $\pi/4$ (C) $\pi/5$ (D) $\pi/6$

Answer and Comments: (D). An excellent example of what an MCQ should *not* be, especially when it is given beforehand that only one of the given options is correct. In the present problem, the hypothesis holds for an equilateral triangle and the answer is $\pi/6$. So, if at all only one option is correct, it has to be (D).

Several honest solutions are possible. A pure geometric solution, contributed by Sagar Sodhani begins by drawing DQ perpendicular to AC and DX parallel to AC . Let DX intersect BE at P as shown in the figure below.



As D is the midpoint of AB and $DP \parallel AE$, P is the midpoint of BE .
So

$$BP = PE \quad (1)$$

Also $DQEP$ is a rectangle and so

$$DQ = PE \quad (2)$$

Using (1) and (2) and the hypothesis $DC = BE$, we get

$$DC = BE = 2PE = 2DQ \quad (3)$$

Therefore $\triangle DQC$ is a right angled triangle in which the hypotenuse DC is twice the side DQ . Hence it is a $90 - 60 - 30$ degrees triangle with $\angle DCQ = \angle DCA = 30^\circ$.

There could hardly be a more elementary proof. It shows the power of ingenious constructions in proofs. The last part could have been expressed using elementary trigonometry. But we have preferred to keep the argument elementary.

With trigonometry, a proof can be given using the sine rule applied to the triangles BEA and CDA both of which have a common angle A . The former is actually a right angled triangle and so simply from the definition of sine, and the hypothesis that D divides AB , we have

$$\sin A = \frac{BE}{AB} = \frac{BE}{2AD} \quad (4)$$

The sine rule applied to $\triangle CDA$ gives

$$\frac{\sin A}{DC} = \frac{\sin \theta}{AD} \quad (5)$$

where $\theta = \angle ACD$. Eliminating $\sin A$ from (4) and (5),

$$\sin \theta = \frac{BE}{2AD} \frac{AD}{DC} = \frac{1}{2} \quad (6)$$

since it is given that $BE = DC$. This solution was contributed by Praveen Agrawal. Yet another solution, contributed by Shubhajit Roy is based on areas. As BE is the altitude, the area, say Δ of $\triangle ABC$ is given by

$$\Delta = \frac{1}{2} BE \cdot AC \quad (7)$$

The area, say Δ_1 of $\triangle ACD$ is given by

$$\Delta_1 = \frac{1}{2} AC \cdot CD \cdot \sin \theta \quad (8)$$

Dividing (8) by (7) and using the hypothesis that $BE = CD$,

$$\sin \theta = \frac{\Delta_1}{\Delta} \quad (9)$$

But this ratio is $\frac{1}{2}$ since a median divides a triangle into two halves of equal areas.

Solutions using trigonometry or areas are shorter than the first solution. This is to be expected since advanced techniques generally save work because a lot of work has already gone into deriving results based on them. (As another example, prove that the sum of the perpendiculars drawn from any point inside an equilateral triangle to the three sides equals its altitude. An elementary solution is lengthy. With areas, the solution is in one line if we divide the triangle into three parts by joining the point to the vertices. This happens because an area is a more advanced concept.)

Vectors are yet another device that sometimes help in solving geometry problems. For example, the shortest proof for the concurrency

of the three altitudes of a triangle is by using dot products of vectors. (See Comment No. 5 of Chapter 8.) For the present problem, a proof using vectors (contributed by Vikash Tiwari) can be given by representing points by their position vectors w.r.t. the vertex C . Referring to the diagram in the statement of the problem, denote the vectors $\vec{CA}$ and $\vec{CD}$ by $\vec{p}$ and $\vec{q}$ respectively. We express the other vectors as linear combinations of $\vec{p}$ and $\vec{q}$ and then use the hypothesis. (The choice of taking these as the basic vectors is dictated by the fact that the angle we want to determine is included in them.) Thus, $\vec{AD} = \vec{CD} - \vec{CA} = \vec{q} - \vec{p}$ and hence $\vec{AB} = 2(\vec{q} - \vec{p})$. This finally gives the position vector of B as

$$\begin{aligned}\vec{CB} &= \vec{CA} + \vec{AB} \\ &= 2\vec{q} - \vec{p}\end{aligned}\tag{10}$$

Also $\vec{CE} = \lambda\vec{p}$ for some λ . Hence

$$\begin{aligned}\vec{BE} &= \lambda\vec{p} - (2\vec{q} - \vec{p}) \\ &= (1 + \lambda)\vec{p} - 2\vec{q}\end{aligned}\tag{11}$$

Perpendicularity of BE and AC gives

$$[(1 + \lambda)\vec{p} - 2\vec{q}] \cdot \vec{p} = 0\tag{12}$$

which simplifies to

$$\vec{p} \cdot \vec{q} = \frac{(1 + \lambda)}{2} \|\vec{p}\|^2\tag{13}$$

On the other hand, the equality of the lengths of $\vec{CD}$ and $\vec{BE}$ translates into

$$\begin{aligned}\|\vec{q}\|^2 &= \|(1 + \lambda)\vec{p} - 2\vec{q}\|^2 \\ &= (1 + \lambda)^2 \|\vec{p}\|^2 + 4\|\vec{q}\|^2 - 4(1 + \lambda)\vec{p} \cdot \vec{q}\end{aligned}\tag{14}$$

(13) and (14) together give

$$3\|\vec{q}\|^2 = (1 + \lambda)^2 \|\vec{p}\|^2\tag{15}$$

Taking square roots,

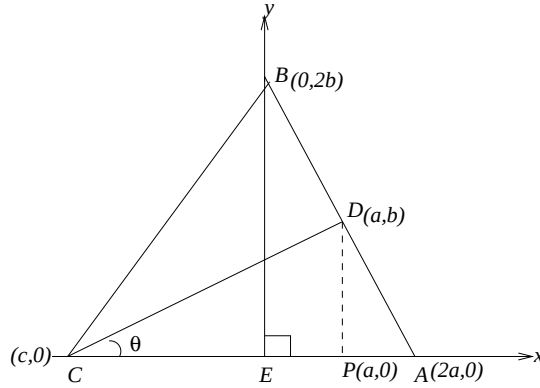
$$\|\vec{q}\| = \frac{1 + \lambda}{\sqrt{3}} \|\vec{p}\| \quad (16)$$

We now have all the information to get hold of the angle $\theta = \angle ACD$.

$$\begin{aligned} \cos \theta &= \frac{\vec{p} \cdot \vec{q}}{\|\vec{p}\| \|\vec{q}\|} \\ &= \frac{(1 + \lambda) \|\vec{p}\|}{2 \|\vec{q}\|} \\ &= \frac{(1 + \lambda) \|\vec{p}\|}{2(1 + \lambda) \|\vec{p}\| / \sqrt{3}} = \frac{\sqrt{3}}{2} \end{aligned} \quad (17)$$

Hence $\theta = \cos^{-1}(\frac{\sqrt{3}}{2}) = \frac{\pi}{6}$.

Finally, we give a solution using coordinates, also contributed by Vikash Tiwari. The calculations are considerably simplified with a judicious choice of the frame of reference. If two lines are perpendicular either from the hypothesis or by some known result, then taking the axes along these lines helps. (See, for example, the Main Problem of Chapter 8 where the axes were taken along the internal and the external angle bisector of an angle of a triangle.) In the present problem, AE is an altitude. So we take E as the origin and one of the axes along BE and the other along the side AC . To highlight this, we draw a different (but equivalent) diagram. For notational simplicity we take A as $(2a, 0)$, B as $(0, 2b)$ and C as $(c, 0)$. Let P be the foot of the perpendicular from D to AC .



With this choice, $D = (a, b)$. Equality of BE and CD gives $(a - c)^2 + b^2 = 4b^2$, i.e.

$$a - c = \sqrt{3}b \quad (18)$$

Hence $\tan \theta = \frac{b}{a-c} = \frac{1}{\sqrt{3}}$ which shows that $\theta = \pi/6$.

Q.11 The number of elements in the set

$$\{x : 0 \leq x \leq 2, |x - x^5| = |x^5 - x^6|\}$$

is

$$(A) 2 \quad (B) 3 \quad (C) 4 \quad (D) 5$$

Answer and Comments: (B). Since $|a|$ can be a or $-a$, the given equality resolves into two possibilities:

$$x - x^5 = x^6 - x^5 \quad (1)$$

$$\text{or } x - x^5 = x^5 - x^6 \quad (2)$$

(1) simplifies to $x^6 - x = 0$ and hence to $x(x-1)(x^4 + x^3 + x^2 + x + 1)$. As the last factor is positive for $x > 0$, there are only two solutions for (1), viz $x = 0$ and $x = 1$.

(2) is a bit more complicated. It simplifies to $x^6 - 2x^5 + x = 0$ Call $x^6 - 2x^5 + x$ as $p(x)$. Again, 0 and 1 are obvious roots of $p(x)$. But they are already included. To see if there are any other roots in $[0, 2]$, we factorise $p(x)$ partially as $xf(x)$ where

$$f(x) = x^5 - 2x^4 + 1 \quad (3)$$

Even though 1 is already a known root of $f(x)$, it is not easy to determine the others. Fortunately, that is not required either. As we only need to find *how many* roots there are, rather than identifying them explicitly, it suffices to study the increasing/decreasing behaviour of $f(x)$. Differentiating,

$$f'(x) = 5x^4 - 8x^3 = x^3(5x - 8) \quad (4)$$

The first factor is positive for all $x > 0$. From the sign of the second factor, we see that $f(x)$ is strictly decreasing on $[0, \frac{8}{5}]$ and strictly increasing on $[\frac{8}{5}, 2]$. Hence $f(x)$ is minimum on $[0, 2]$ at $\frac{8}{5}$. Moreover,

$f(\frac{8}{5}) < 0$ as $f(1) = 0$ and f is decreasing on $[0, \frac{8}{5}]$. On the other hand, $f(2) = 1$. Hence by the Intermediate Value Property, $f(x)$ has at least one root in $(\frac{8}{5}, 2)$. Moreover this root is unique since f is strictly increasing on $[\frac{8}{5}, 2]$.

All put together, the given set contains exactly three elements, viz., 0, 1 and the (unidentified but unique) root of $f(x)$ in $(\frac{8}{5}, 2)$.

Finding the set of solutions of (1) is trivial. The real problem is to find the number of solutions of (2). Doing so directly by studying the polynomial $p(x) = x^6 - 2x^5 + x$ is not easy. Factorisation is the only way. The tricky part is the partial factorisation (3). It is tempting to think that a total factorisation as $x(x-1)g(x)$ where $g(x) = x^4 - x^3 - x^2 - x - 1$ will make things simpler. But this turns out not to be the case. Finding the roots of $g(x)$ is not easy and determining its increasing/decreasing behaviour is also not as easy as of $f(x)$ because $g'(x)$ does not factor as simply as $f'(x)$. It is tempting to find the number of roots of $g(x)$ by drawing a graph of $g(x)$. But to draw a revealing graph, you have to put in a lot of work in the first place. A good graph does save time, but often only when drawn by someone else!

Normally we think that the more you factorise a polynomial, the easier it is to study its roots. The present problem is unusual in that here partial factorisation works better than a more elaborate factorisation. This paradoxical situation arises because here we are not identifying the roots *per se* of $p(x)$ but only their locations. This is done by determining the increasing/decreasing behaviour of the relevant factor. For this we need derivatives and it so happens that the sign determination of the derivative of the fifth degree factor $x^5 - 2x^4 + 1$ is much easier than that of the derivative of the fourth degree factor $x^4 - x^3 - x^2 - x - 1$.

A discerning candidate will notice that we have shown that $p(x)$ has three roots. But a real polynomial of degree 6 has 6 roots. Even though two of them may be complex, there still has to be one more real root. Where is it? The answer lies in the fact that our interest was in finding the number of roots in $[0, 2]$. The reasoning we have given actually shows that even the bigger interval $[0, \infty)$ contains only 3 roots. The missing root is, therefore, in $(-\infty, 0)$.

Q.12 Let $n \geq 1$. The maximum possible number of primes in the set $\{n+6, n+7, \dots, n+35\}$ is

(A) 7 (B) 8 (C) 12 (D) 13

Answer and Comments: (B). The given set is a set, say S_n , of 30 consecutive integers beginning with $n + 6$ and the problem asks to find the maximum number of primes it can have. The primes get rarer and rarer as you go along the number line. It can, in fact, be shown that there exist arbitrarily large sets of consecutive integers with no primes. For example, for every positive integer k , the set of $k - 1$ integers from $k! + 2$ to $k! + k$ can contain no prime because $k! + r$ is divisible by r for every $r = 2, 3, \dots, k$. So there is no point in asking the minimum number of primes in S_n .

Anyway coming to the problem, in $S_1 = \{7, 8, \dots, 35, 36\}$, the primes are 7, 11, 13, 17, 19, 23, 29 and 31. So the answer is at least 8. We claim that it cannot be more than 8.

To show that no S_n contains 9 (or more) primes, we note that every prime larger than 3 is either of the form $6k + 1$ or of the form $6k - 1$ for some positive integer k . In a set with 30 consecutive integers, there are exactly 5 integers of the form $6k + 1$, say, $6k_1 + 1, 6k_2 + 1, 6k_3 + 1, 6k_4 + 1$ and $6k_5 + 1$. Moreover, these integers, $k_1, k_2, \dots, k_5$ themselves form a consecutive block. (For example, in the set $\{101, 102, \dots, 130\}$, the integers 103, 109, 115, 121 and 127 are of the form $6k + 1$ for $k = 17, 18, 19, 20$ and 21 respectively.) Now, every block of 5 consecutive integers contains exactly one integer congruent to 4 modulo 5. If this integer is r , then $6r + 1$ is divisible by 5 and hence not a prime. (In the illustration just given, $r = 19$ and correspondingly, $6r + 1 = 115$.) Thus out of the five possible candidates for primality at least one is not a prime. Some other candidates, e.g. 121 in the illustration, may disqualify because of some other reason. But in any case, we conclude that S_n can have at most 4 primes of the form $6n + 1$. By an analogous reasoning, it can have at most 4 primes of the form $6n - 1$. Put together, 8 is an upper bound on the number of primes in S_n for $n \geq 1$.

The problem could have been made more testing by giving 10 as a possible option. For, those who rely only on the fact that all primes greater than 3 are of the form $6k \pm 1$ are likely to think that 10 is the maximum if they fail to observe divisibility by 5 of two of these candidates.

However, 10 is indeed the correct maximum if the block size were 36 rather than 30. A simple proof is to divide the block into two parts, the first of size 30 and the second of the tail of 6 elements. The first part can contain at most 8 primes and the second at most 2. That the maximum 10 is actually reached is proved, for example, by the block $\{7, 8, \dots, 42\}$ in which, apart from 7, 11, 13, 17, 19, 23, 29 and 31 already listed, 37 and 41 are also primes.

- Q.13 Suppose 40 distinguishable balls are to be distributed into 4 different boxes such that each box gets exactly 10 balls. Out of these 40 balls, 10 are defective and 30 are non-defective. In how many ways can the balls be distributed such that the defective balls go to the first two boxes?

$$(A) \frac{40!}{(10!)^4} \quad (B) \frac{30!.20!}{(10!)^5} \quad (C) \frac{20!.20!}{(10!)^5} \quad (D) \frac{30!.10!}{(10!)^4}$$

Answer and Comments: (B). Call the boxes as B_1, B_2, B_3 and B_4 . The 20 non-defective that must go to $B_3 \cup B_4$ can be chosen from the 30 non-defective balls in $\binom{30}{20}$ ways. And for each choice the ten balls that go to B_3 can be chosen in $\binom{20}{10}$ ways. So, m , the number of possible ways to fill B_3 and B_4 is

$$m = \frac{30!}{20!10!} \times \frac{20!}{10!10!} = \frac{30!}{(10!)^3} \quad (1)$$

For each such placement, the remaining 10 non-defective balls, along with the 10 defective ones must go to $B_1 \cup B_2$. Every such placement will have r non-defective and $10-r$ defective balls going to B_1 for some integer r from 0 to 10. After this, there is no choice left for B_2 . Hence, the number, say n , of such placements is

$$n = \sum_{r=0}^{10} \binom{10}{r} \binom{10}{10-r} \quad (2)$$

$$= \binom{20}{10} \quad (3)$$

$$= \frac{20!}{(10!)^2} \quad (4)$$

where, in going from (2) to (3) we have used the binomial theorem for the coefficient of x^{10} in the expansion of $(1+x)^{10}(1+x)^{10}$.

The answer mn comes by multiplying (1) and (4).

A simple combinatorial problem.

Q.14 The number of positive solutions to the equation

$$e^x \sin x = \log x + e^{\sqrt{x}} + 2$$

is

$$(A) 0 \quad (B) 1 \quad (C) 2 \quad (D) \infty$$

Answer and Comments: (D). This problem resembles Q.11 in spirit. But there the two functions to be compared were polynomials and hence the number of points of equality was necessarily finite. Here neither side is a polynomial in x and so the number of solutions could be infinite.

But once again, our task is not to identify the solutions but only to count how many of them are there. So, analysis of the increasing/decreasing behaviour of the two functions and an application of the Intermediate Value Property seem to be the right tools.

Call the functions on the left and right sides as $f(x)$ and $g(x)$. $g(x)$ is strictly increasing on $(0, \infty)$ with range $(-\infty, \infty)$. Moreover, $g(x) > 0$ for all $x > 1$ as all three terms are positive. As $x \rightarrow \infty$, even though both $\log x$ and $e^{\sqrt{x}}$ tend to ∞ , the latter does so much faster in the sense that the ratio $\frac{\log x}{e^{\sqrt{x}}}$ tends to 0. The same holds for the ratio $\frac{2}{e^{\sqrt{x}}}$. Hence there exists some $M > 1$ such that for all $x > M$

$$0 < e^{\sqrt{x}} < g(x) < 2e^{\sqrt{x}} \quad (1)$$

As for $f(x)$, the factor e^x is positive for all $x > 0$ and larger than $2e^{\sqrt{x}}$ for all sufficiently large x . However, $\sin x$ can be close to 0, in fact, even negative. Still for every positive integer k , we have

$$f(x) = e^x \quad (2)$$

for $x = (4k+1)\pi/2$. and

$$f(x) = -e^x < 0 \quad (3)$$

for $x = (4k + 3)\pi/2$.

Fix some $M > 0$ such that (1) holds for all $x > M$. Now let k be any integer for which $(4k + 1)\pi/2$ (and hence $(4k + 3)\pi/2$ too) exceed M . Then from (3),

$$f(x) < 0 < g(x) \quad (4)$$

for $x = (4k + 3)\pi/2$, while, combining (1) and (2),

$$0 < g(x) < f(x) \quad (5)$$

for $x = (4k + 1)\pi/2$. Let $h(x) = f(x) - g(x)$. Then from (5), $h((4k + 1)\pi/2) > 0$ while from (4), $h((4k + 3)\pi/2) < 0$. As h is continuous, by the IVP, there is some $x \in ((4k + 1)\pi/2, (4k + 3)\pi/2)$ at which $h(x) = 0$, i.e. $f(x) = g(x)$. So, this interval contains at least one solution of the given equation. For distinct k 's, the corresponding intervals are disjoint and so the solutions are distinct. As there are infinitely many integers k which satisfy $(4k + 1)\pi/2 > M$, the given equation has infinitely many solutions.

A good problem based on the periodicity of the sine function and the relative rates of growth of the functions $e^x, e^{\sqrt{x}}$ and $\log x$. The argument could also be given graphically. The crucial point is that the graph of $f(x)$ oscillates between the graphs e^x and $-e^x$ and the amplitudes of oscillations increase indefinitely.

- Q.15 In a room with $n \geq 2$ people, each pair shakes hands between themselves with probability $\frac{2}{n^2}$ and independently of other pairs. If p_n is the probability that the total number of handshakes is at most 1, then $\lim_{n \rightarrow \infty} p_n$ is equal to

(A) 0 (B) 1 (C) e^{-1} (D) $2e^{-1}$

.

Answer and Comments: (D). There are two parts to the question, the calculation of p_n for each $n \geq 2$ and then the evaluation of the limit of the sequence $\{p_n\}$ as $n \rightarrow \infty$.

For the first part, the correct sample space is not the set of n people in the room, but rather, the set, say S , of all (unordered) pairs of these

people. Clearly there are $\binom{n}{2} = \frac{n(n-1)}{2}$ such pairs. For each pair, the probability of a handshake is $\frac{2}{n^2}$ and the complementary probability of not shaking hands is $1 - \frac{2}{n^2}$. (The case $n = 1$ gives a negative probability. But it is excluded by the stipulation $n \geq 2$.)

The desired probability p_n is the sum of the probabilities of two mutually disjoint events, (i) no pair shakes hands and (ii) exactly one pair shakes hands. Both are binomial probabilities. The first event has probability $(1 - \frac{2}{n^2})^{n(n-1)/2}$ while the probability of the second event is $\frac{n(n-1)}{2} \times \frac{2}{n^2} \times (1 - \frac{2}{n^2})^{\frac{n(n-1)}{2}-1}$. Let us call these as a_n and b_n respectively. Then

$$p_n = a_n + b_n \quad (1)$$

To find $\lim_{n \rightarrow \infty} p_n$, we evaluate $\lim_{n \rightarrow \infty} a_n$ and $\lim_{n \rightarrow \infty} b_n$ separately and add. (There is a possibility that neither a_n nor b_n tends to a limit but $a_n + b_n$ does. But we shall first try the simpler approach.)

Even though the expression for b_n is more complicated than for a_n , its limit is easier to find because the factor $\frac{n(n-1)}{2} \times \frac{2}{n^2}$ tends to 1 as $n \rightarrow \infty$, being the ratio of two monic polynomials of degree 2. Hence calling the remaining factor as c_n ,

$$\lim_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} (1 - \frac{2}{n^2})^{\frac{n(n-1)}{2}-1} \quad (2)$$

As the terms of sequence on the R.H.S. are powers where both the base and the exponent vary with n , it is convenient to take logarithms before taking the limit.

$$\ln c_n = \ln(1 - \frac{2}{n^2})^{\frac{n(n-1)}{2}-1} = (\frac{n(n-1)}{2} - 1) \ln(1 - \frac{2}{n^2}) \quad (3)$$

If we divide the second factor by $\frac{2}{n^2}$, its limit becomes familiar. So we multiply and divide by $\frac{2}{n^2}$ to get

$$\ln c_n = \frac{2}{n^2} \times \frac{n^2 - n - 2}{2} \times \frac{\ln(1 - \frac{2}{n^2})}{\frac{2}{n^2}} \quad (4)$$

Once again, as $n \rightarrow \infty$, the product of the first two factors tends to 1. Hence

$$\lim_{n \rightarrow \infty} \ln c_n = \lim_{n \rightarrow \infty} \frac{\ln(1 - \frac{2}{n^2})}{\frac{2}{n^2}} \quad (5)$$

The last limit is a familiar one, with value -1 . (Those who are not familiar with it may replace the discrete variable $\frac{2}{n^2}$ by the continuous variable x and evaluate $\lim_{x \rightarrow 0^+} \frac{\ln(1-x)}{x}$ using either the l'Hôpital's rule or directly from the definition of the derivative of the function $\ln(1-x)$.)

Thus we have shown that $\ln c_n$ tends to -1 as $n \rightarrow \infty$. Taking exponentials, c_n and hence b_n tends to e^{-1} as $n \rightarrow \infty$.

We still have to find $\lim_{n \rightarrow \infty} a_n$. The method adopted is the same. We consider $\ln a_n = \frac{n(n-1)}{2} \ln(1 - \frac{2}{n^2})$, multiply and divide by $\frac{2}{n^2}$ and use the same familiar limit again to get that $\lim_{n \rightarrow \infty} \ln a_n = -1$. So a_n also tends to e^{-1} as $n \rightarrow \infty$. Adding, we finally get that $p_n = a_n + b_n$ tends to $2e^{-1}$ as $n \rightarrow \infty$. ■

The question is a disjoint combination of two problems, the calculation of a probability and evaluation of limits of sequences. There is some duplication of work in the second part. The problem would have been more interesting if there were some way to calculate the limit of p_n without evaluating it explicitly.

Q.16 Let $P = \{(x, y) : x+1 \geq y, x \geq -1, y \geq 2x\}$. Then the minimum value of $(x+y)$ where (x, y) varies over the set P is

- (A) -1 (B) -3 (C) 3 (D) 0

Answer and Comments: (B). This is a problem of minimisation of a function of two variables over a subset of $\mathbb{R}^2$. This is beyond the HSC syllabus. But in the present case the function to be minimised, viz. $x+y$ is linear and P is a polygonal region. So this comes under a very special case of optimisation, called **linear programming** which, nowadays, is covered by most HSC boards. The basic result is that the maximum as well as the minimum occurs at a vertex of the region. As a polygon has only finitely many vertices, a straight comparison of the values of the function to be optimised (called the **objective function**) gives the answer.

In the present problem, P is a triangle bounded by the straight lines

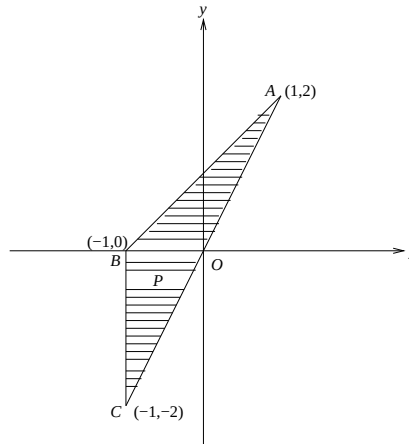
$$y = x + 1 \tag{1}$$

$$x = -1 \quad (2)$$

$$\text{and } y = 2x \quad (3)$$

The vertices of P can be determined by solving these equations simultaneously two at a time. (1) and (2) together give $(-1, 0)$ as a vertex, (2) and (3) give $(-1, -2)$ as another vertex. The remaining vertex is $(1, 2)$ which comes from solving (1) and (3) together. The values of $x + y$ at these vertices are, respectively, -1 , -3 and 3 respectively. Clearly the minimum is -3 . ■

Those not familiar with this basic result (viz. that the optimum occurs only at a vertex) can proceed by directly sketching the triangle P . Each of the three inequalities gives a half plane and P is the intersection of these three half planes. It is shown by shading in the figure below.



The existence of a point, say $Q_0 = (x_0, y_0)$, where f is minimum on P is a matter of theoretical calculus in several variables which is beyond HSC level. But once we accept it, showing that Q_0 must be the vertex C is not difficult. To begin with, Q_0 cannot be an interior point of P . For if it is, then draw a sufficiently small disc of radius r centred at Q_0 and lying inside P . Then the point $Q_1 = (x_0 - \frac{r}{2}, y_0 - \frac{r}{2})$ also lies in P , but $f(Q_1) = x_0 + y_0 - r < x_0 + y_0 = f(Q_0)$, contradicting the minimality of $f(Q_0)$. Since Q_0 cannot be an interior point of P , it has to be a boundary point. The search is now reduced to the three sides

of the triangle P . On each side, we can express $f(x, y) = x + y$ as a function of a single variable. For example, the equation of the side AB is $y = x + 1$. Hence $f(x, y) = x + y = 2x + 1$. Clearly, this decreases as x decreases and so the minimum of f on the segment AB occurs at the end point B . By a similar reasoning, its minimum on any side occurs at one of the end points. Hence it suffices to narrow the search for the minimum to the set of the vertices of P .

The same logic applies if the region P is a polygon with n sides where $n \geq 3$. The equations of the lines containing the sides are given. But the trouble is that not every pair determines a vertex of P because some of such points of intersection may lie outside P while some of them may lie in the interior of P . In such a case, the vertices have to be identified by sketching P . When the number of variables is 3, the situation is far more complicated because we have to consider the points of intersection of all triads from the given set of planes. There is a systematic algorithm called the **simplex method** to go from one vertex to an adjacent vertex where the objective function has a lower (or higher, depending on the interest) value till no such vertex exists.

Q.17 Let $n > 1$ be the smallest composite integer that is coprime to $\frac{10000!}{9900!}$. Then

- (A) $n \leq 100$ (B) $100 < n \leq 9900$
 (C) $9900 < n \leq 10000$ (D) $n > 10000$

Answer and Comments: (D). A composite integer is one which is not a prime. In other words, if n is a composite integer, then it has a (positive) proper divisor m , i.e. a divisor different from 1 and n itself. Then by definition, $n = mm'$ where $m' = \frac{n}{m}$ is an integer and also a proper divisor of n . Out of m and m' the smaller one is less than $\sqrt{n}$. In case n is a perfect square and $m = \sqrt{n}$, then $m' = m$. In either case, we conclude that every composite integer n has a proper divisor not exceeding $\sqrt{n}$.

Now consider the number $\frac{10000!}{9900!}$. Call it N . Then

$$N = 9901.9902. \dots 9998.9999.10000 \quad (1)$$

We claim that N is divisible by $100!$ and hence by every integer from 1 to 100. The easiest way to see this is to consider the binomial coefficient $\binom{10000}{100}$ which is, after all, a whole number to say the least. Hence $\frac{10000!}{9900!}$ equals $100!$ times this binomial coefficient.

Now suppose n is a composite number coprime to N . By the first observation above, n has a factor, say m , which is at most $\sqrt{n}$. This factor m cannot be between 2 and 100 as otherwise it will divide N too and then N and n will not be coprime. So $m > 100$ and hence $n \geq m^2 > 10000$. So the smallest composite number which is coprime to N is bigger than 10000. ■

An excellent problem in number theory requiring only elementary knowledge of divisibility and binomial coefficients.

The first observation made above improves the efficiency of the most elementary method of testing primality of a given positive integer n . A brute force method would be to check the divisibility of n by all the integers m from 2 to $n-1$ one after another until either we find a proper divisor or until we exhaust all possible divisors. Some improvement arises by noting that once it is tested that n is not divisible by some $k < n$, there is no need to check its divisibility by multiples of k . A more substantial improvement comes from the observation made above, because it shows that we only need to carry the divisibility check by m till $m \leq \sqrt{n}$. Since $\sqrt{n}$ is much smaller than n , this is a considerable saving. For example, to test whether 9997 is a prime, we only need to check if it is divisible by integers from 2 to 99 since $100 > \sqrt{9997}$.

Q.18 The real number x satisfies

$$\frac{|x|^2 - |x| - 2}{2|x| - |x|^2 - 2} > 2$$

if and only if x belongs to

- (A) $(-2, -1) \cup (1, 2)$ (B) $(-2/3, 0) \cup (0, 2/3)$
 (C) $(-1, -2/3) \cup (2/3, 1)$ (D) $(-1, 0) \cup (0, 1)$

Answer and Comments: (C). The problem asks the solution set of a certain inequality involving x . It involves only $|x|$. Since $|x| = |-x|$, we may as well find the solution set for $x > 0$. Then that for $x < 0$ will be its reflection in the point 0. So, replacing $|x|$ by x throughout, the given inequality becomes

$$\frac{x^2 - x - 2}{2x - x^2 - 2} > 2 \quad (1)$$

To find the solution set (for $x > 0$), it is tempting to rewrite (1) as

$$x^2 - x - 2 > 2(2x - x^2 - 2) \quad (2)$$

But this would be valid only for those x for which the factor $2x - x^2 - 2$ is positive. Rewriting it as $-1 - (x - 1)^2$ we see that it is negative for all x . Hence the inequality in (2) has to be reversed. So, for $x > 0$, we want the solution set of

$$x^2 - x - 2 < 2(2x - x^2 - 2) \quad (3)$$

which simplifies to

$$3x^2 - 5x + 2 < 0 \quad (4)$$

The L.H.S. is a quadratic in x with positive leading coefficient and roots $\frac{2}{3}$ and 1. So the expression is negative in between these two roots, i.e. when $x \in (2/3, 1)$.

So the solution set of the given inequality for $x > 0$ is $(2/3, 1)$. As already said, the solution set for $x < 0$ would be its mirror image, i.e. the interval $(-1, -2/3)$. Put together, x satisfies the given inequality if and only if $x \in (-1, -2/3) \cup (2/3, 1)$. ■

In all the options, the solution set is symmetric around the origin. Perhaps this is meant to give a hint that it suffices to find the solution set only for $x > 0$. The real catch is the sign determination of the expression in the denominator of the given inequality. The problem is simple once you cross that hurdle.

- Q.19 Let $A = \{1, \dots, 5\}$ and $B = \{1, \dots, 10\}$. Then the number of ordered pairs (f, g) of functions $f : A \rightarrow B$ and $g : B \rightarrow A$ satisfying $(g \circ f)(a) = a$ for all $a \in A$ is

(A) $\frac{10!}{5!} \times 5^5$ (B) $5^{10} \times 5!$ (C) $10! \times 5!$ (D) $\binom{10}{5} \times 10^5$

Answer and Comments: (A). Let R be the range of the function f . Then R is a subset of B and has at most 5 elements. Since $g(f(a)) = a$ for all $a \in A$, f is injective and so R has exactly 5 elements. Once R is determined, there are $5!$ functions f which map A onto R because each such function amounts to a permutation of the 5 elements of R . Also values of g on elements of R are already determined by f . But the values of g on the remaining 5 elements of B are completely unrestricted and can be chosen in 5^5 ways. Since R can be chosen in $\binom{10}{5}$ ways, the total number of ordered pairs of functions (f, g) with the given restrictions is $\binom{10}{5} \times 5! \times 5^5$, which, upon simplification becomes $\frac{10!}{5!} 5^5$. ■

A simple counting problem once it is realised that f has a left inverse and hence is one-to-one.

Q.20 Let

$$S = \frac{1}{\sqrt{10000}} + \frac{1}{\sqrt{10001}} + \dots + \frac{1}{\sqrt{160000}}$$

Then the largest positive integer not exceeding S is

(A) 200 (B) 400 (C) 600 (D) 800

Answer and Comments: (C). In simpler language, we are asked to find the integral part, commonly denoted by $[S]$ of the given sum S . The number of terms is prohibitively large to add one by one. Nor is there any trick in the nature of the terms that will simplify the sum drastically. (A popular example of such a trick is where the k -th terms from both the ends add to something independent of k . See Q.16 of Paper 2 of JEE 2020.)

To add to the huge number of terms, most of them are irrational. So we can evaluate them only approximately in terms of decimals. And this has to be done in a such a way that the total error is less than 1 (as the problem amounts to place S in an interval of length 1 with integer end points). So the approximations of the terms individually will have to be of a very high degree of precision.

Fortunately, the first and the last terms of S are very simple, $\frac{1}{100}$ and $\frac{1}{400}$ respectively. It turns out that if we drop any one of these we get a sum which can be related to a certain integral, which can be evaluated very easily. Letting $f(x) = \frac{1}{\sqrt{x}}$,

$$S - \frac{1}{100} = \sum_{x=10001}^{x=160000} f(x) \quad (1)$$

$$\text{and } S - \frac{1}{400} = \sum_{x=10000}^{x=159999} f(x) \quad (2)$$

Let us look at these sums in a novel way. Consider the interval $[10000, 160000]$ and partition it into 150000 subintervals each having length 1. The nodes of this partition are precisely the integers from 10001 to 159999. As $f(x)$ is strictly decreasing on each subinterval of the partition, its maximum occurs at the left end point and minimum at the right end point. Therefore, if we multiply each term of these sums by 1 (the length of the subinterval), then they become the lower and the upper Riemann sums of $f(x)$ over this partition.

Call the definite integral $\int_{10000}^{160000} f(x)dx$ as I . It always lies between the lower and the upper Riemann sums. Hence we have

$$S - \frac{1}{100} < I < S - \frac{1}{400} \quad (3)$$

and hence

$$I + \frac{1}{400} < S < I + \frac{1}{100} \quad (4)$$

It remains to calculate I . This is easy.

$$I = \int_{10000}^{160000} \frac{dx}{\sqrt{x}} = 2\sqrt{x} \Big|_{10000}^{160000} = 2(400 - 100) = 600 \quad (5)$$

Putting this into (4),

$$600 + \frac{1}{400} < S < 600 + \frac{1}{100} \quad (6)$$

As the first and the last term have the same integral part, viz. 600, so does the middle term. ■

The question is strikingly similar to the second part of Exercise (17.10). See also Q.9 of Paper 1 of JEE 2008. Normally, when an integral cannot be evaluated exactly (which is generally the case in integrals appearing in real life problems in contrast to integrals asked in examinations), a Riemann sum, or often a suitable linear combination of several Riemann sums w.r.t. a sufficiently fine partition of the domain interval, is used to approximate the integral. This leads to some well-known rules like the **trapezoidal rule** or the **Simpson's rule**. In the present problem, the tables are turned around. That is, integrals are used to approximate certain sums.

Q.21 The limit

$$\lim_{n \rightarrow \infty} \frac{2 \log 2 + 3 \log 3 + \dots + n \log n}{n^2 \log n}$$

is equal to

(A) 0 (B) 1/4 (C) 1/2 (D) 1

Answer and Comments: (C). Yet another problem based on the relationship between an integral and a Riemann sum. Call the sum in the numerator as S_n . Let $f(x) = x \log x$ for $x \geq 1$, Then $f(x)$ is strictly increasing and S_n is its upper Riemann sum for the interval $[1, n]$ divided into $n - 1$ equal parts. So, if we let

$$I_n = \int_1^n x \log x dx \quad (1)$$

Then we have

$$I_n \leq S_n \quad (2)$$

The lower Riemann sum for $f(x)$ for the same partition is $1 \log 1 + 2 \log 2 + (n - 1) \log(n - 1)$ which is simply $S_n - n \log n$. Hence we have

$$S_n - n \log n \leq I_n \leq S_n \quad (3)$$

Dividing throughout by $n^2 \log n$ (which is positive for all $n > 1$), we have

$$\frac{S_n}{n^2 \log n} - \frac{1}{n} \leq \frac{I_n}{n^2 \log n} \leq \frac{S_n}{n^2 \log n} \quad (4)$$

By the Sandwich theorem, the desired limit equals $\lim_{n \rightarrow \infty} \frac{I_n}{n^2 \log n}$. To finish the solution, we need to evaluate I_n . This can be done using integration by parts.

$$\begin{aligned} I_n &= \int_1^n x \log x dx \\ &= \left(\frac{x^2}{2} \log x - \frac{x^2}{4} \right) \Big|_1^n \\ &= \frac{n^2 \log n}{2} - \frac{n^2}{4} + \frac{1}{4} \end{aligned} \tag{5}$$

When divided by $n^2 \log n$, the first term is a constant $\frac{1}{2}$ while the other two terms tend to 0 as $n \rightarrow \infty$. Therefore the given limit is $\frac{1}{2}$. ■

There is considerable conceptual duplication with the last problem. It could have been avoided.

Q.22 The set of all real numbers x for which $3^{2^{1-x^2}}$ is an integer has

- (A) 3 elements (B) 15 elements
- (C) 24 elements (D) infinitely many elements

Answer and Comments: (B). The expression $3^{2^{1-x^2}}$, as a function of y is rather complicated to handle, because the exponent of 3, viz. 2^{1-x^2} is itself an exponential, with exponent $1 - x^2$. And even in it there is an exponential x^2 . So we have a chain of three functions each involving an exponential.

It is convenient to introduce an intermediate variable. Let

$$y = 2^{1-x^2} \tag{1}$$

We want those values of y for which 3^y is an integer. There is no dearth of such values. As a function of y , 3^y is a strictly increasing function which maps $(-\infty, \infty)$ onto $(0, \infty)$. So its range includes all positive integers. Given any positive integer r , there is a unique y for which $3^y = r$. In fact, y is given explicitly by $y = \log_3 r$. (This is in fact, the very definition of logarithms. It is as informative as saying that Mrs. X is Mr. X's wife.)

As there are infinitely many positive integers, it is tempting to think that (D) is the correct option. But there is a catch, even though given a positive integer r there is some $y \in \mathbb{R}$ for which $3^y = r$, we also want that this y be of the form 2^{1-x^2} for some real number x . In other words, y should be in the range of another function $f(x)$ defined by

$$f(x) = 2^{1-x^2} \quad (2)$$

To find the range of $f(x)$ we must first find the range of $1 - x^2$. This is a very familiar function with range $(-\infty, 1]$. As with powers of 3, those of 2 also increase strictly as the exponent does. So, the range of $f(x)$ is $(0, 2^1]$, i.e. $(0, 2]$.

So, y can only vary in $(0, 2]$. Correspondingly, 3^y can vary only in $(3^0, 3^2]$, i.e. in $(1, 9]$. This interval contains 8 integers, those from 2 to 9 and correspondingly,

$$y = \log_3 k, \quad k = 2, 3, 4, 5, 6, 7, 8, 9 \quad (3)$$

For each of these values, we have to find how many real numbers there are which satisfy

$$2^{1-x^2} = \log_3 k \quad (4)$$

or equivalently,

$$1 - x^2 = \log_2(\log_3 k) \quad (5)$$

It is possible to solve this explicitly as

$$x = \pm \sqrt{1 - \log_2(\log_3 k)} \quad (6)$$

For the 7 integers k from 2 to 8, the expression under the radical sign is positive and so there are two square roots. However, for $k = 9$, the R.H.S. is 0 and so $x = 0$ is the only solution. Together the number of solutions is $2 \times 7 + 1 = 15$. ■

The introduction of the language of a set at the start of the problem is quite unnecessary. It would have been fine to simply ask for the number of real values of x for which $3^{2^{1-x^2}}$ is an integer. Barring this

minor stylistic point, the problem is very well designed and an excellent test of not only the knowledge of logarithms, but of the ability to think analytically. One of the options could have been 16 elements in order to penalise those candidates who miss the exceptionality of the value 9 of y . But perhaps the paper setters thought it is unfair to penalise someone with good analytical thinking for a trivial lapse.

- Q.23 Let $n > 1$, and let us arrange the expansion of $(x^{1/2} + \frac{1}{2x^{1/4}})^n$ in decreasing powers of x . Suppose the first three coefficients are in arithmetic progression. Then, the number of terms where x appears with an integer power, is

(A) 3 (B) 2 (C) 1 (D) 0

Answer and Comments: (A). As a notational simplification, we put $y = x^{1/4}$. This will avoid having to work with fractional exponents. The expression to be expanded becomes $(y^2 + \frac{1}{2y})^n$ and the goal now is to find the number of terms in the expansion in which the exponent of y is divisible by 4.

The first three terms of the expansion are y^{2n} , $\binom{n}{1}y^{2(n-1)}\frac{1}{2y}$ and $\binom{n}{2}y^{2(n-2)}\frac{1}{4y^2}$. The coefficients in them are $1, \frac{n}{2}$ and $\frac{n(n-1)}{8}$ respectively. These are given to form an A.P. That gives us an equation for n , viz.

$$n = 1 + \frac{n^2 - n}{8} \quad (1)$$

On simplification, this reduces to a quadratic equation

$$n^2 - 9n + 8 = 0 \quad (2)$$

which has two solutions, $n = 1$ and $n = 8$. The first one is excluded by hypothesis. So we have $n = 8$.

To check in which terms in the expansion of $(y^2 + \frac{1}{2y})^n$, y appears with an exponent which is a multiple of 4, we list these powers in decreasing order, dropping the coefficients as they are not relevant. Every time we move to the next term, the exponent to which y^2 is raised goes down by 1 while in the denominator (of the second term in the parentheses) the exponent to which y is raised goes up by 1. The net effect is that the exponent of y goes down by 3. This tallies with the exponents $2n, 2n-3$

and $2n - 6$ of the first three terms which we already obtained. So, the next exponents of y will be $2n - 9, 2n - 12, 2n - 15, \dots$ and so on. Putting $n = 8$, the exponents of the nine terms in the expansion are, 16, 13, 10, 7, 4, 1, -2 , -5 and -8 respectively. Only three of these, viz., 16, 4 and -8 are divisible by 4. ■

A simple but good problem on binomial expansion.

Q.24 Consider points of the form (n, n^k) , where n and k are integers with $n \geq 0, k \geq 1$. How many such points are strictly inside the circle of radius 10 with centre at the origin?

(A) 11 (B) 12 (C) 15 (D) 17

Answer and Comments: (A). Stripped of the geometrical garb, the problem amounts to asking how many ordered pairs of the given form (n, n^k) are there with

$$n^2 + n^{2k} < 100 \quad (1)$$

As $k \geq 1$, $n^k \geq n$ and so the L.H.S. is at least $2n^2$. So we only need to consider the cases where $n^2 < 50$. That gives 0 to 7 as the possible values of n . For each of these we list the possibilities.

(0,0)
 (1,1)
 (2,2), (2,4), (2,8)
 (3,3), (3,9)
 (4,4)
 (5,5)
 (6,6)
 (7,7)

In all there are 11 points. ■

An absolutely trivial counting problem.

Q.25 Let $p < q$ be prime numbers such that $p^2 + q^2 + 7pq$ is a perfect square. Then the largest possible value of q is

(A) 7 (B) 11 (C) 23 (D) 29

Answer and Comments: (B). Let

$$r^2 = p^2 + q^2 + 7pq \quad (1)$$

Then $r^2 - (p + q)^2 = 5pq$. Factoring,

$$(r + p + q)(r - p - q) = 5pq \quad (2)$$

These two equations are, in general, insufficient to determine the values of the unknowns p, q and r . But here the variables have integer values and moreover at least two of them, viz. p and q are primes. 5 is also a prime. Hence the R.H.S. is a product of three primes and can be factored in $2^3 = 8$ possible ways. In each such factorisation, the larger factor would be $r + p + q$ and the smaller one $r - p - q$. When we are unable to decide which factor is larger, we have to consider both the cases.

Let

$$a = r + p + q \quad (3)$$

$$\text{and } b = r - p - q \quad (4)$$

Eliminating r , we get

$$a - b = 2(p + q) \quad (5)$$

We begin by listing all possible factorisations of $5pq$ as $a \times b$ and in each case see if there is any solution.

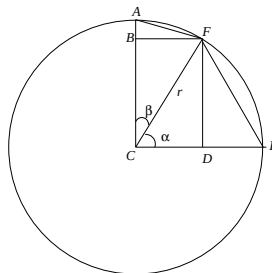
- (i) $a = 5pq, b = 1$. Then (5) gives $2p + 2q = 5pq - 1$ which is impossible since the L.H.S. is less than $4q$ while the R.H.S. exceeds $4q$ (as $p \geq 2$).
- (ii) $a = 5p, b = q$. Here $2p + 2q = 5p - q$ which implies $p = q$ contradiction. The other possibility $a = q, b = 5p$ gives $q - 5p = 2p + 2q$ and hence $7p = -q$, a negative number.
- (iii) $a = 5q, b = p$. This gives $5q - p = 2p + 2q$ which again gives $p = q$. The other possibility $a = p, b = 5q$ cannot hold since $q > p$.

- (iv) $a = pq, b = 5$. Here $2p + 2q = pq - 5$. This can be rewritten as $pq - 2p - 2q + 4 = 9$, i.e. $(p - 2)(q - 2) = 9$. Since the factor $(p - 2)$ is smaller than $(q - 2)$, we must have $p - 2 = 1, q - 2 = 9$, whence $p = 3, q = 11$. This does lead to a solution because $3^2 + 11^2 + 7 \times 3 \times 11 = 9 + 121 + 231 = 361 = 19^2$. The possibility $a = 5, b = pq$ need not be considered since obviously $pq > 5$ because the smallest primes are 2 and 3.

The question asks for the largest possible value of q . We have, in fact, shown that the *only* possible value of q is 11. ■

A very good problem requiring only elementary properties of primes. To avoid confusion, the question should have simply asked the value of q .

- Q.26 In the adjoining figure, C is the centre of the circle drawn. A, F, E lie on the circle and $BCDF$ is a rectangle. If $\frac{DE}{AB} = 2$, then $\frac{FE}{FA}$ equals



- (A) $\sqrt{\frac{3}{2}}$ (B) $\sqrt{2}$ (C) $\sqrt{\frac{5}{2}}$ (D) $\sqrt{3}$

Answer and Comments: (B). We can introduce coordinates, taking CE and CA as axes. But a trigonometric approach is better because there is a handy formula for the length of a chord of a circle in terms of the angle it subtends at the centre. So, let α, β be the angles $\angle FCE$ and $\angle FCB$ respectively. Let r be the radius of the circle. (Actually, as only ratios are involved, we may assume that the radius is 1. But that's not much of a saving.) Then

$$DE = CE - CD = r - r \cos \alpha = 2r \sin^2 \frac{\alpha}{2} \quad (1)$$

and similarly,

$$AB = AC - BC = r - r \cos \beta = 2r \sin^2 \frac{\beta}{2} \quad (2)$$

As we are given that $\frac{DE}{AB} = 2$, we have

$$\frac{\sin^2 \frac{\alpha}{2}}{\sin^2 \frac{\beta}{2}} = 2 \quad (3)$$

By the formula for the length of a chord,

$$\frac{FE}{FA} = \frac{2r \sin \frac{\alpha}{2}}{2r \sin \frac{\beta}{2}} \quad (4)$$

From (3), this ratio equals $\sqrt{2}$. ■

A simple application of trigonometry. An alert candidate will notice that the angles α and β add to $\frac{\pi}{2}$. But that is nowhere needed.)

Q.27 For every increasing function $b : [1, \infty) \rightarrow [1, \infty)$ such that

$$\int_1^\infty \frac{dx}{b(x)} < \infty,$$

we must have

$$\begin{array}{ll} \text{(A)} \sum_{k=1}^{\infty} \frac{\sqrt{\log k}}{b(k)} < \infty & \text{(B)} \sum_{k=1}^{\infty} \frac{\log k}{b(\log k)} < \infty \\ \text{(C)} \sum_{k=1}^{\infty} \frac{e^k}{b(e^k)} < \infty & \text{(D)} \sum_{k=3}^{\infty} \frac{1}{\sqrt{b(\log k)}} < \infty \end{array}$$

Answer and Comments: (C). Nothing is given about the function $b(x)$ other than that it is positive (in fact, always at least 1), increasing and the integral $\int_1^\infty \frac{dx}{b(x)}$ is finite. It is precisely with these hypothesis that we get the well known Maclaurin Integral Test which relates the convergence of an improper integral with that of an infinite series of positive terms. Specifically, the test says that

$$\lim_{k \rightarrow \infty} \int_1^k \frac{1}{b(x)} < \infty \text{ if and only if } \sum_{k=1}^{\infty} \frac{1}{b(k)} \text{ is convergent} \quad (1)$$

The basic idea of the proof is merely that since $\frac{1}{b(x)}$ is a decreasing function, for any $1 \leq c < d$

$$\frac{1}{b(c)}(d - c) \geq \int_c^d \frac{dx}{b(x)} \geq \frac{1}{b(d)}(d - c) \quad (2)$$

(1) is proved by applying this double inequality to unit intervals starting at $1, 2, \dots, k-1$. Adding these $k-1$ inequalities and using the additivity property of the integrals gives

$$\sum_{r=1}^{k-1} \frac{1}{b(r)} \geq \int_1^k \frac{dx}{b(x)} \geq \sum_{r=2}^k \frac{1}{b(r)} \quad (3)$$

Hence the convergence of the series $\sum_{k=1}^{\infty} \frac{1}{b(k)}$ and that of the improper integral $\int_1^{\infty} \frac{dx}{b(x)}$ imply each other.

We follow a very similar approach in proving (C), except that now we are already given the convergence of the integral $\int_1^{\infty} \frac{dx}{b(x)}$ and our goal is to show that the series $\sum_{k=1}^{\infty} \frac{e^k}{b(e^k)}$ is convergent. We can rewrite this as $\sum_{k=1}^{\infty} \frac{1}{e^{-k}b(e^k)}$. If we make a substitution $x = e^u$, then

$$\begin{aligned} \int_1^{\infty} \frac{dx}{b(x)} &= \int_0^{\infty} \frac{e^u}{b(e^u)} du \\ &= \int_0^{\infty} \frac{du}{e^{-u}b(e^u)} \end{aligned} \quad (4)$$

Convergence of the integral in (4) implies that of the integral $\int_1^{\infty} \frac{du}{e^{-u}b(u)}$ (which is only (4) minus a constant $\int_0^1 \frac{du}{e^{-u}b(u)}$) and hence of $\int_1^{\infty} \frac{dx}{e^{-x}b(x)}$ by changing the dummy variable of integration. If we call $e^{-x}b(x)$ as $B(x)$ then we now know that the integral $\int_1^{\infty} \frac{dx}{B(x)}$ is convergent. It

is tempting to apply (1) now to conclude that $\sum_{k=1}^{\infty} \frac{1}{B(k)}$ is convergent.

Since that is the same series as $\sum_{k=1}^{\infty} \frac{e^k}{b(e^k)}$ we are done.

But there is a catch in this argument. We are given that $b(x)$ is increasing. But the factor e^{-x} decreases (and very rapidly too) as x increases. So there is no guarantee that the product $B(x) = e^{-x}b(x)$ is increasing. (That will happen if $b(x)$ increases faster than e^x . But that need not always be the case, e.g. when $b(x) = x^2$.)

So, we try a different approach which is based not directly on (1) but on its derivation. We shall use (2) again. But instead of applying it to the intervals of the form $[r, r+1]$ for $r = 1, 2, \dots, k$, we shall apply it to intervals of the form $[e^{r-1}, e^r]$ for $r = 1, 2, \dots, k-1$. Moreover, unlike (1) which is a two way statement, our interest now is only in proving the convergence of a series from the already given convergence of the integral $\int_1^{\infty} \frac{dx}{b(x)}$. So we take only the right half of (2) and get

$$\int_{e^{r-1}}^{e^r} \frac{dx}{b(x)} \geq \frac{e^r - e^{r-1}}{b(e^r)} \quad (5)$$

for $r = 1, 2, \dots, k-1$. As before we add these $k-1$ inequalities and use the additivity property of the integrals to get

$$\begin{aligned} \int_1^{e^k} \frac{dx}{b(x)} &\geq \sum_{r=1}^k \frac{e^r - e^{r-1}}{b(e^r)} \\ &= \sum_{r=1}^k \frac{e^r}{b(e^r)} - \frac{1}{e} \sum_{r=1}^k e^r b(e^r) \\ &= \frac{e-1}{e} \sum_{r=1}^k \frac{e^r}{b(e^r)} \end{aligned} \quad (6)$$

Multiplying both the sides by $\frac{e}{e-1}$ and letting $k \rightarrow \infty$, we get

$$\frac{e}{e-1} \int_1^{\infty} \frac{dx}{b(x)} \geq \sum_{k=1}^{\infty} \frac{e^k}{b(e^k)} \quad (7)$$

This proves the convergence of the series $\sum_{k=1}^{\infty} \frac{e^k}{b(e^k)}$ since its terms are all positive and all partial sums are bounded. ■

Problems based on the MacLaurin integral test are not frequently asked in HSC level examinations. The present problem requires not the test in its familiar form but the reasoning in its proof. So this is an excellent problem for Paper 2. By asking in Paper 1, those who take the short cut of applying the MacLaurin test without checking the monotonicity of the integrand go scot free.

Q.28 Let $\psi : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with $\int_{-1}^1 \psi(x)dx = 1$. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function. Then

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \int_{1-\epsilon}^{1+\epsilon} f(y)\psi\left(\frac{1-y}{\epsilon}\right)dy$$

equals

- (A) $f(1)$ (B) $f(1)\psi(0)$
 (C) $f'(1)\psi(0)$ (D) $f(1)\psi(1)$

Answer and Comments: (A). Call the integral in the expression as I_ϵ . The substitution $y = 1 - \epsilon t$ transforms it to

$$\begin{aligned} I_\epsilon &= \int_{1-\epsilon}^{1+\epsilon} f(y)\psi\left(\frac{1-y}{\epsilon}\right)dy \\ &= -\epsilon \int_1^{-1} f(1 - \epsilon t)\psi(t)dt \\ &= \epsilon \int_{-1}^1 f(1 - \epsilon t)\psi(t)dt \end{aligned} \tag{1}$$

Hence, the limit to be evaluated, say L , becomes

$$L = \lim_{\epsilon \rightarrow 0} \int_{-1}^1 f(1 - \epsilon t)\psi(t)dt \tag{2}$$

Here t is the variable of integration and ϵ is a parameter. As the parameter varies, so does the integral and we have to find its limit as $\epsilon \rightarrow 0$. It is tempting to do so by bringing the limit inside the integral. If we do so, the integrand tends to $f(1)\psi(t)$ and we can take $f(1)$ outside the integral as it is a constant independent of t . So the answer would be $f(1)\int_{-1}^1 \psi(t)dt$ which equals $f(1)$.

But an interchange of integration and limits is not valid in general. What can go wrong is easier to see if instead of a parameter ϵ tending to 0 through real values, we have a discrete parameter n which tends to ∞ through integer values. If we have a sequence $\{f_n(x)\}_{n \in \mathbb{N}}$ all defined over an interval $[a, b]$ and $f(x)$ is a function such that for every $x \in [a, b]$, $f_n(x) \rightarrow f(x)$, then in general we cannot say that $\int_a^b f_n(x)dx \rightarrow \int_a^b f(x)dx$ as $n \rightarrow \infty$.

To evaluate the limit L , we use the differentiability of f and integration by parts. First we define

$$\theta(t) = \int_{-1}^t \psi(t)dt \quad (3)$$

for $t \in [-1, 1]$. Then $\theta(-1) = 0$ and $\theta(1) = 1$ by hypothesis. θ is a function defined by integrating ψ and so because of continuity of ψ and the fundamental theorem of calculus,

$$\theta'(t) = \psi(t) \quad (4)$$

for all $t \in [-1, 1]$.

To evaluate the integral in (2), we apply (4) and integration by parts.

$$\begin{aligned} \int_{-1}^1 f(1 - \epsilon t) \psi(t) dt &= \int_{-1}^1 f(1 - \epsilon t) \theta'(t) dt \\ &= (f(1 - \epsilon t) \theta(t)) \Big|_{-1}^1 + \epsilon \int_{-1}^1 f'(1 - \epsilon t) \theta(t) dt \end{aligned}$$

We need not evaluate the integral in the last term. For, regardless of its value, the term will tend to 0 as $\epsilon \rightarrow 0$ because of the factor ϵ in front. Hence

$$\begin{aligned} L &= \lim_{\epsilon \rightarrow 0} f(1 - \epsilon) \theta(1) - f(1 - \epsilon) \theta(-1) \\ &= f(1) \theta(1) - f(1) \theta(-1) \\ &= f(1) \end{aligned} \quad (5)$$

since $\theta(1) = 1$ and $\theta(-1) = 0$ as noted already. ■

We have found the same answer as earlier, but legitimately. The question fits more appropriately in Paper 2, as otherwise those who take the unwarranted short cut will get rewarded.

Q.29 Let a, b, c be three complex numbers. The equation

$$az + b\bar{z} + c = 0$$

represents a straight line on the complex plane if and only if

- (A) $a = b$ (B) $\bar{a}c = b\bar{c}$
 (C) $|a| = |b| \neq 0$ (D) $|a| = |b| \neq 0$ and $\bar{a}c = b\bar{c}$

Answer and Comments: (D). In real coordinates, the equation of a straight line is of the form

$$px + qy + r = 0 \tag{1}$$

with the stipulation that $p^2 + q^2 \neq 0$ as otherwise the equation will represent either the empty set or the whole plane depending on whether $r \neq 0$ or $r = 0$. Multiplying by 2 and writing $2x$ as $z + \bar{z}$ and $2y$ as $(z - \bar{z})/i = -i(z - \bar{z})$ this becomes

$$(p - iq)z + (p + iq)\bar{z} + 2r = 0 \tag{2}$$

or equivalently,

$$\bar{u}z + u\bar{z} + 2r = 0 \tag{3}$$

where $u = p + iq$ with $|u| \neq 0$.

Verbally, the coefficients of z and $\bar{z}$ are non-zero and complex conjugates of each other and the constant term is real. Now to find the conditions under which the given equation represents a straight line, we make two cases, depending on whether $c = 0$ or $c \neq 0$.

If $c = 0$, then the constant term is already real and the part $\bar{a}c = b\bar{c}$ is satisfied. Further a, b are non-zero and complex conjugates of each other, which means that the first part of (D) is also satisfied.

In case c in the given equation is non-zero, we multiply it throughout by $\bar{c}$ and get

$$a\bar{c}z + b\bar{c}\bar{z} + |c|^2 = 0 \tag{4}$$

Now the constant term is real and the equation will represent a straight line if and only if $a\bar{c}$ and $b\bar{c}$ are non-zero and complex conjugates of each other, i.e.

$$\bar{a}c = b\bar{c} \tag{5}$$

This implies $|a||c| = |b||c| \neq 0$ and hence $|a| = |b|$ since $|c| \neq 0$.

Thus in both the cases, (D) is a necessary and sufficient condition for the given equation to represent a straight line. ■

Q.30 Consider the following two statements:

- (I) There exists a differentiable function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(x^3 + x^5) = e^x - 100$.
- (II) There exists a continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(e^x) = x^3 + x^5$.

Then

- (A) Only (I) is correct.
- (B) Only (II) is correct.
- (C) Both (I) and (II) are correct.
- (D) Neither (I) nor (II) is correct.

Answer and Comments: (D). If (I) is true, then differentiating both the sides, we get

$$g'(x^3 + x^5)(3x^2 + 5x^4) = e^x \quad (1)$$

for all x . But the L.H.S. vanishes at $x = 0$ while the R.H.S. is always positive, a contradiction. Hence (I) is false.

For (II), if we put $t = \ln x$ for $x > 0$, then the equation implies

$$g(x) = g(e^{\ln t}) = (\ln t)^3 + (\ln t)^5 \quad (2)$$

For $x \leq 0$, we cannot put $t = \ln x$ and so the equation above does not hold. But $g(0)$ has to be defined as some number. Moreover, if g is to be continuous, then, in particular, the right handed limit of $g(x)$ at 0 must exist and equal $g(0)$. But from the equation above as $x \rightarrow 0^+$, $\ln t \rightarrow -\infty$ and hence the R.H.S. also tends to $-\infty$ which is not a real number. So there is no way to define $g(0)$ so as to make $g(x)$ continuous. Hence (II) is false too. ■

ISI BStat-BMath-UGB-2024 Paper 2 with Comments

Q.1 Find, with proof, all possible values of t such that

$$\lim_{n \rightarrow \infty} \left\{ \frac{1 + 2^{1/3} + 3^{1/3} + \dots + n^{1/3}}{n^t} \right\} = c$$

for some real number $c > 0$. Also find the corresponding values of c .

Solution and Comments: The problem is strikingly similar to Exercise (17.11), except that there the exponent of every term in the numerator was 10. It can be proved algebraically that for every positive integer r , the sum $1^r + 2^r + \dots + n^r$ is a polynomial of degree $r+1$ with leading coefficient $\frac{1}{r+1}$. (The cases $r = 1, 2, 3$ where the sums equal $\frac{n(n+1)}{2}$, $\frac{n(n+1)(2n+1)}{6}$ and $\frac{n^2(n+1)^2}{4}$ respectively are very familiar.) So in order for the ratio to tend to some number, the highest power of n in the denominator must be n^{r+1} .

In the present problem, the exponents of all the terms in the numerator are $\frac{1}{3}$. So we expect that t should be $\frac{4}{3}$. In Paper 1, this guess would be rewarding. But now we have to give a reasoning.

If we divide the numerator as well as the denominator by $n^{1/3}$ the expression, say E , becomes

$$E = \frac{\left(\frac{1}{n}\right)^{1/3} + \left(\frac{2}{n}\right)^{1/3} + \dots + \left(\frac{n}{n}\right)^{1/3}}{n^{t-\frac{1}{3}}} \quad (1)$$

If we further multiply both the numerator and the denominator by $\frac{1}{n}$, we get

$$E = \frac{\left[\left(\frac{1}{n}\right)^{1/3} + \left(\frac{2}{n}\right)^{1/3} + \dots + \left(\frac{n}{n}\right)^{1/3}\right] \times \frac{1}{n}}{n^{t-\frac{4}{3}}} \quad (2)$$

The numerator now is precisely the upper Riemann sum for the function $f(x) = x^{1/3}$ for the interval $[0, 1]$ partitioned into n equal parts. As $n \rightarrow \infty$, it will tend to the integral $\int_0^1 x^{1/3} dx$. By an easy calculation, this integral is $\frac{3}{4}$.

So, for the ratio to tend to a definite limit, the (new) denominator, must tend to some constant as n tends to ∞ . This is possible only for

$t = \frac{4}{3}$. So our guess is correct. When $t = \frac{4}{3}$, the denominator is simply 1. As the limit of the numerator is $4/3$, this is the corresponding value of c . ■

A good, but rather common, application of Riemann integrals in evaluating limits of certain sums by recognising them as Riemann sums. Q.20 and Q.21 of Paper 1 are of a similar spirit. Thus there is not a duplication, but a triplication of ideas.

Q.2 Suppose $n \geq 2$. Consider the polynomial

$$Q_n(x) = 1 - x^n - (1 - x)^n.$$

Show that the equation $Q_n(x) = 0$ has only two real roots, namely, 0 and 1.

Solution and Comments: By a direct verification, both 0 and 1 are roots of $Q(x)$. If there were some other root, say α , then by Rolle's theorem, $Q'(x)$ would have at least two distinct roots in between these three roots. But

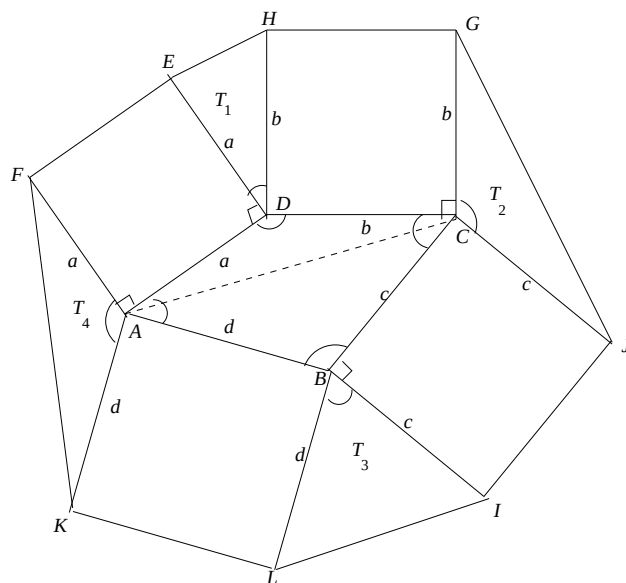
$$Q'(x) = -nx^{n-1} + n(1 - x)^{n-1} \quad (1)$$

Hence for $Q'(x)$ to vanish we must have $x^{n-1} = (1 - x)^{n-1}$. If n is even, then $n - 1$ is odd and the only possibility is $x = 1 - x$, i.e. $x = \frac{1}{2}$. If n is odd, $n - 1$ is even. But the possibility $x = -(1 - x)$ has no solution. So, whether n is even or odd, $Q'(x)$ has only one root, and accordingly, $Q(x)$ has only two roots, 0 and 1. ■

A very simple problem once the idea of applying Rolle's theorem strikes.

Q.3 Let $ABCD$ be a quadrilateral with all internal angles $< \pi$. Squares $ADEF, DCGH, CBIJ, BAKL$ are drawn on each side as shown in the figure below. Let T_1, T_2, T_3, T_4 be the triangles EDH, GCJ, IBL, KEF respectively. Let Δ_i be the area of T_i for $i = 1, 2, 3, 4$. Prove that

$$\Delta_1 - \Delta_2 + \Delta_3 - \Delta_4 = 0.$$



Solution and Comments: Possibly because of limitations on drawing figures in printing, in old mathematics books, the statements of problems used to be purely verbal and certain words were coined to convey what was intended. Thus squares drawn outwards would be ‘escribed’. Also notations were often dispensed with using words. Thus the triangles T_1, T_2, T_3, T_4 would be described as triangles with vertices at the vertices of the quadrilateral and adjacent sides as the sides of the escribed squares on sides adjacent to the vertices of the quadrilateral. Until you got used to such a language, it would take a while to understand what is meant.

Anyway, coming to the problem, consider the triangle T_1 . Of the four angles which have D as a vertex, two are right angles. So the remaining two add up to π . Hence $\angle EDH$ of T_1 has the same sine as $\angle D$ of the quadrilateral. Also its adjacent sides are a and b . Hence

$$\Delta_1 = \frac{1}{2}ab \sin \angle D \quad (1)$$

Similarly,

$$\Delta_3 = \frac{1}{2}cd \sin \angle B \quad (2)$$

If we draw the diagonal AC , we see that the right hand sides of the equations above are also the areas of $\triangle ADC$ and $\triangle ABC$ respectively. Together they add to the area of the quadrilateral $ABCD$. A similar argument shows that Δ_2 and Δ_4 add to the area of the quadrilateral. Hence

$$\Delta_1 + \Delta_3 = \Delta_2 + \Delta_4 \quad (3)$$

each side equaling the area of the quadrilateral $ABCD$. Therefore $\Delta_1 - \Delta_2 + \Delta_3 - \Delta_4 = 0$ as was to be proved. ■

Q.4 Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be a function which is differentiable at 0. Define another function $g : \mathbb{R} \longrightarrow \mathbb{R}$ as follows:

$$g(x) = \begin{cases} f(x) \sin(\frac{1}{x}) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Suppose that g is also differentiable at 0. Prove that

$$g'(0) = f'(0) = f(0) = g(0) = 0.$$

Solution and Comments: Already g is given to vanish at 0. From continuity of $g(x)$ at 0, $g(x) \rightarrow 0$ as $x \rightarrow 0$. From the definition of $g(x)$, this means

$$\lim_{x \rightarrow 0} f(x) \sin(\frac{1}{x}) = 0 \quad (1)$$

From this we claim that $f(x) \rightarrow 0$ as $x \rightarrow 0$. By continuity of f at 0, $\lim_{x \rightarrow 0} f(x)$ exists and equals $f(0)$. Call this limit as L . If $L \neq 0$, then $f(x) \sin(\frac{1}{x})$ tends to L as x tends to 0 through values of the form $\frac{2}{(4n+1)\pi}, n \in \mathbb{N}$ but to 0 as $x \rightarrow 0$ through values of the form $\frac{1}{n\pi}$ for $n \in \mathbb{N}$, a contradiction. So we must have $L = 0$ and hence $f(0) = 0$.

So far, we have shown $g(0) = 0$ (which is actually given) and $f(0) = 0$ using only continuity of g and f . But we are given something stronger, viz. that both of them are differentiable. From that we have to show that both f' and g' vanish at 0. Here too, from the differentiability of

$g(x)$ we first prove that $f'(0)$. Since $g(0) = 0 = f(0)$,

$$\begin{aligned} g'(0) &= \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0} \\ &= \lim_{x \rightarrow 0} \frac{f(x)}{x} \sin\left(\frac{1}{x}\right) \\ &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \sin\left(\frac{1}{x}\right) \end{aligned} \quad (2)$$

We cannot rewrite this hastily as

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} \times \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) \quad (3)$$

because the second limit does not exist. But the first one definitely exists and equals $f'(0)$. Call this limit as M . If $M = 0$, as otherwise, by the same reasoning as before, $\frac{g(x)-g(0)}{x-0}$ will tend to different limits as $x \rightarrow 0$ through different sets of values, contradicting existence of $g'(0)$. So $M = 0$ and therefore $g'(0) = 0$ too. During the course of the proof, we have already proved that $f'(0) = 0$ because $f'(0) = M$. Thus we have proved that all the numbers $f'(0)$, $g'(0)$ and $f(0)$ vanish. ■

The function $\sin(\frac{1}{x})$ tends to no definite limit as $x \rightarrow 0$. But its boundedness masks this demerit when multiplied by some other function such as x (or some other power of x). As a result, functions of the form $x^m \sin(\frac{1}{x})$ are stock sources of problems of checking smoothness conditions such as continuity, differentiability, continuous differentiability and so on. The unusual feature of the present problem is that the function $f(x)$ is not given but some smoothness conditions about it and its product with $\sin(\frac{1}{x})$ are given and we are asked to prove that certain numbers vanish. The problem itself, although not difficult, is an excellent test of the logical order in which to proceed. So it fits most appropriately in Paper 2.

Q.5 Let $P(x)$ be a polynomial with real coefficients. Let $\alpha_1, \alpha_2, \dots, \alpha_k$ be the distinct real roots of $P(x) = 0$. If P' is the derivative of P , show that for each $i = 1, 2, \dots, k$,

$$\lim_{x \rightarrow \alpha_i} \frac{(x - \alpha_i)P'(x)}{P(x)} = r_i,$$

for some positive integer r_i .

Solution and Comments: The problem asks to prove not only that a certain limit is a positive integer, but indicates that this limit may be different for different roots. Thereby the paper setters have given a hint that the answer lies in some integer parameter associated with a root of a polynomial.

By far, the simplest, and also arguably the most commonly used integer parameter associated with a root of a polynomial is its multiplicity. If α is a root of a polynomial $P(x)$, then $x - \alpha$ is a factor of $P(x)$. Sometimes some higher power of $(x - \alpha)$ may also divide $P(x)$. The **multiplicity** of α as a root of $P(x)$ is defined as the highest positive integer r for which $(x - \alpha)^r$ divides $P(x)$. Equivalently, it is the unique integer r for which we can write $P(x)$ as $(x - \alpha)^r Q(x)$ where $Q(x)$ is a polynomial which does not have α as a root, i.e. $Q(\alpha) \neq 0$.

Now assume r_i is the multiplicity of α_i as a root of $P(x)$. Then we can write

$$P(x) = (x - \alpha_i)^{r_i} Q_i(x) \quad (1)$$

where $Q_i(x)$ is a polynomial with $Q_i(\alpha_i) \neq 0$

Differentiating,

$$P'(x) = (x - \alpha_i)^{r_i} Q'_i(x) + r_i (x - \alpha_i)^{r_i-1} Q_i(\alpha_i) \quad (2)$$

Multiplying (2) throughout by $(x - \alpha_i)$ and dividing by (1) gives

$$\frac{(x - \alpha_i)P'(x)}{P(x)} = (x - \alpha_i)Q'_i(x) + r_i \quad (3)$$

As $x \rightarrow \alpha_i$, the R.H.S. tends to r_i . That proves the assertion. ■

The concept of multiplicity can be extended, under a different name, to other functions and their zeros. But if α a zero of $f(x)$, then dividing $f(x)$ by $x - \alpha$ makes no sense as $f(x)$ may not be a polynomial. For example, π is a zero of $\sin x$. But we cannot write $\frac{\sin x}{x - \pi}$ as a function which is defined at $x = \pi$. However, we can take the limit of this ratio as $x \rightarrow \alpha$. It is simply $\cos \pi = -1 \neq 0$. In such a case we say that π is

a zero of **order** 1 (or a simple zero) of $\sin x$. In general, a number α is said to be a zero of order r of a function $f(x)$, if $\frac{f(x)}{(x - \alpha)^r}$ tends to a finite non-zero limit as $x \rightarrow \alpha$. Accordingly, 0 is a zero of order 2 of the function $f(x) = 1 - \cos x$. For sufficiently differentiable functions, there is yet another description of the order, viz., it is the smallest integer r for which $f^{(r)}(\alpha)$, i.e. the r -th derivative of $f(x)$ at α is non-zero.

The problem amounts to asking to show that the multiplicity of a root of a polynomial is the same as its order.

- Q.6 Let $x_1, x_2, \dots, x_{2024}$ be non-negative real numbers with $\sum_{i=1}^{2024} x_i = 1$. Find, with a proof, the minimum and maximum possible values of the expression

$$\sum_{i=1}^{1012} x_i + \sum_{i=1}^{2024} x_i^2.$$

Solution and Comments: There is, of course, nothing special about the figure 2024. Following a practice started by IMO many years ago, it has now become a cliché for annual competitive examinations to include a problem which contains the year in which it is held as a numerical parameter, regardless of its relevance to the mathematical content.

The wording of the problem is somewhat ambiguous. To stress that the elements appear in a particular order, it would have been better to say that $(x_1, x_2, \dots, x_{2024})$ is a finite sequence of real numbers in $[0, 1]$ whose terms add to 1. Among all such possible sequences, we have to find the maximum and the minimum values of the given expression, say E .

As there is nothing special about 2024, we could replace it with any even integer. The simplest case is when $n = 2$. Then with a slight change of notation, the problem asks to find the minimum and maximum of $x + y^2$ where $x, y \in [0, 1]$ and $x + y = 1$. This is an easy problem in calculus, where we replace y by $1 - x$ and consider the function $f(x) = x + (1 - x)^2 = x^2 - x + 1 = (x - \frac{1}{2})^2 + \frac{3}{4}$ whose maximum is 1 and occurs at the end points 0, 1 and minimum $\frac{3}{4}$ occurring at $\frac{1}{2}$. The situation with more terms is more complicated because now

the problem is to optimise a function of more than two variables, well beyond the HSC level.

Nevertheless, we see from this miniature version that the maximum occurs at an end point and the minimum at an interior point.

Coming to the given problem, we begin rewriting the expression, say E , to be optimised slightly. Because of the constraint that the 2024 terms add to 1, we have

$$\sum_{i=1}^{1012} x_i + \sum_{i=1013}^{2024} x_i = 1 \quad (1)$$

Hence,

$$E = \sum_{i=1}^{1012} x_i + \sum_{i=1013}^{2024} x_i^2 \quad (2)$$

$$= 1 + \sum_{i=1013}^{2024} (x_i^2 - x_i) \quad (3)$$

Since $0 \leq x_i \leq 1$, $x_i^2 \leq x_i$ with equality holding only for $x_i = 0$ and $x_i = 1$. Hence no term in the summation in (3) is positive. So the maximum of the R.H.S. is at most 1. To show that the value 1 is indeed attained, it suffices to take any one x_i as 1 and all the remaining x_j 's as 0. Therefore, the maximum, say M , of the expression E is 1. (Of courses, it is also attained in many other ways, e.g. by choosing the first 1012 terms so that they add to 1 and setting the rest of the terms to 0 each.)

Calculating the minimum, say m , of E is a little more involved. The minimum of every term in the sum in (3) is $-\frac{1}{4}$ occurring when $x_i = \frac{1}{2}$. But obviously not all terms can equal $\frac{1}{2}$. The constraint (1) along with non-negativity of each x_i puts a severe restriction on how many terms can be $\frac{1}{2}$. (In the miniature version above where there were only two terms instead of 2024, the minimum did occur when both the terms were $\frac{1}{2}$ each. But this cannot happen when the number of terms is large. And that's what calls for a new approach.)

To find when the minimum occurs, let us take a closer look at the expression for E in (3). On the face of it, it appears that it depends only on the second half of the terms (i.e. the terms from x_{1013} to x_{2024})

and the first half has no role. But that is not quite true because the two halves are related by (1). Nevertheless, we observe that if some number $x \in (0, 1)$ appears as one of the terms in the first half, then from (2) its contribution to E is x . But if the same number appears in the second half, its contribution x^2 is smaller. So probably, for E to be minimum, the terms in the first half should be negligibly small. Ideally, they should all be 0.

Surprisingly this guess turns out to be correct, that is, when E is minimum we claim that

$$x_1 = x_2 = \dots = x_{1012} = 0 \quad (4)$$

To prove this, suppose

$$p = x_1 + x_2 + \dots + x_{1012} \quad (5)$$

Verbally, p is the sum of the terms in the first half. p is some number between 0 and 1. For every such fixed p , let m_p be the minimum possible value of E in which (5) holds. Then, the minimum m of E we are looking for will be given by

$$m = \min\{m_p : 0 \leq p \leq 1\} \quad (6)$$

To calculate m_p , we need to minimise the contribution from terms in the second half. In other words, we must minimise

$$x_{1013}^2 + x_{1014}^2 + \dots + x_{2024}^2 \quad (7)$$

subject to the constraint that

$$x_{1013} + x_{1014} + \dots + x_{2024} = 1 - p \quad (8)$$

For this we apply the Cauchy-Schwarz inequality to the sequences, $(x_{1013}, x_{1014}, \dots, x_{2024})$ and $(1, 1, \dots, 1)$ each of length 1012. The inequality gives

$$\begin{aligned} (x_{1013} \cdot 1 + x_{1014} \cdot 1 + \dots + x_{2024} \cdot 1)^2 &\leq \left(\sum_{i=1013}^{2024} x_i^2 \right) \times \sum_{i=1013}^{2024} 1^2 \\ &= \left(\sum_{i=1013}^{2024} x_i^2 \right) \times 1012 \end{aligned} \quad (9)$$

Because of (8), this becomes

$$\left(\sum_{i=1013}^{2024} x_i^2 \right) \geq \frac{(1-p)^2}{1012} \quad (10)$$

with equality holding if and only if

$$x_{1013} = x_{1014} = \dots = x_{2024} = \frac{1-p}{1012} \quad (11)$$

So, the minimum possible value m_p for a given $p \in [0, 1]$ is

$$m_p = p + \frac{(1-p)^2}{1012} \quad (12)$$

$$\begin{aligned} &= p + \frac{1-2p+p^2}{1012} \\ &= \frac{1+1010p+p^2}{1012} \end{aligned} \quad (13)$$

Clearly, m_p increases as p increases, with minimum occurring at $p = 0$. This confirms our guess that for E to be minimum, all terms from x_1 to x_{1012} must vanish. Moreover, by (6) we now get the minimum value of E as $\frac{1}{1012}$. Hence the maximum and minimum possible values of E are 1 and $\frac{1}{1012}$ respectively. ■

As mentioned earlier, we could have replaced 2024 by any even integer, say $2n$. In that case, the maximum M would still be the same, viz. 1, occurring when one x_i equals 1 and all others vanish. But the minimum m changes with n , because instead of (12) we shall have

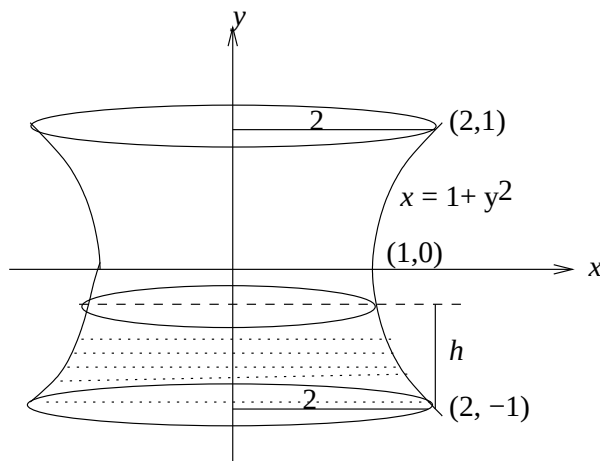
$$\begin{aligned} m_p &= p + \frac{(1-p)^2}{n} \\ &= \frac{1+(n-2)p+p^2}{n} \end{aligned} \quad (14)$$

If $n \geq 2$, m_p will increase as p increases and its minimum m will occur when $p = 0$. Its value will be $\frac{1}{n}$. However, when $n = 1$, $m_p = 1 - p + p^2$ is not an increasing function of p . Its minimum occurs not at $p = 0$ but at $p = \frac{1}{2}$ and the minimum value is $\frac{3}{4}$. This is exactly the miniature

version of the problem we dealt with earlier, where instead of 2024 terms we had only 2 terms. It is ironic that a miniature version, which is generally meant to illustrate a problem, turned out to be exceptional. But this happens sometimes. The smallest prime 2 is exceptional in that it is the only even prime and many theorems about primes fail for it. So it is not an illustrative prime.

An excellent problem on inequalities.

- Q.7 Consider a container of the shape obtained by revolving a segment of the parabola $x = 1 + y^2$ around the y -axis as shown below. The container is initially empty. Water is poured at a constant rate of $1 \text{ cm}^3/\text{s}$ into the container. Let $h(t)$ be the height of water inside the container at time t . Find the time t when the rate of change of $h(t)$ is maximum.



Solution and Comments: A routine problem on volumes enclosed by surfaces of revolution and related rates. Let t denote the time measured in seconds, $h = h(t)$ the height (or depth) of water in cm.s and $V = V(t)$ the volume of the water in the container at time t , measured in cubic cm.s. both functions of t .

Based on our experience of bottle necks, it is intuitively obvious that the height will grow most rapidly when the area of a horizontal cross section is the smallest. But we can also prove this rigorously. A typical

horizontal cross section of the container is a disc of radius $1 + y^2$. This equals 2 when $y = \pm 1$ with $y = -1$ for the bottom and $y = 1$ for the top. Since h is measured from the bottom, we have

$$y = -1 + h \quad (1)$$

Hence the volume of water when the height is h is

$$V = \pi \int_{-1}^{h-1} (1 + y^2)^2 dy \quad (2)$$

We can evaluate this integral and express V as a function of h . But that is not necessary. Our interest is in $\frac{dh}{dt}$ which can be found by differentiating (2) and using the fundamental theorem of calculus. Thus,

$$\begin{aligned} \frac{dV}{dt} &= \frac{dV}{dh} \frac{dh}{dt} \\ &= (1 + (h - 1)^2)^2 \frac{dh}{dt} \end{aligned} \quad (3)$$

The L.H.S. is given to be a constant. So even without expressing h explicitly as a function of t , we see that $\frac{dh}{dt}$ is maximum when $(1 + (h - 1)^2)^2$ is minimum, i.e. when $h = 1$. For this value of h , $y = 0$ and $1 + y^2$ is smallest. So the horizontal cross section by $y = 0$ is smallest, confirming our intuitive prediction.

But the problem does not end here. We also have to find the time it takes to reach this height. And for this we do need to express $h(t)$ as a function of t and then solve the equation $h(t) = 1$ for t . This can be done by setting the L.H.S. of (3) equal to 1 and then solving the resulting differential equation for h , with the initial condition $h(0) = 0$.

But there is a better way. We simply calculate the volume, say V , between $h = 0$ and $h = 1$. As water is being poured at a constant rate, we would know the time it takes. The values of y for $h = 0$ and $h = 1$ are -1 and 0 respectively, we have

$$V = \pi \int_{-1}^0 (1 + y^2)^2 dy \quad (4)$$

(which is hardly different from (2) since we now know that $h = 1$). An easy calculation gives

$$V = \pi \left(y + \frac{2}{3}y^3 + \frac{1}{5}y^5 \right) \Big|_{-1}^0 = \pi \left(1 + \frac{2}{3} + \frac{1}{5} \right) = \frac{28}{15} \pi \text{ cm}^3 \quad (5)$$

As water is poured at the rate of cubic centimeters per second, the time taken to reach the height 1 is $\frac{28}{15}\pi$ seconds. ■

A simple but good problem to test common sense.

- Q.8 In a sports tournament involving N teams, each team plays every other team exactly once. At the end of every match, the winning team gets 1 point and the losing team gets 0 points. At the end of the tournament, the total points received by the individual teams are arranged in decreasing order as follows:

$$x_1 \geq x_2 \geq \dots \geq x_N.$$

Prove that for any $1 \leq k \leq N$,

$$\frac{N-k}{2} \leq x_k \leq N - \frac{k+1}{2}.$$

Solution and Comments: Let T_i be the team which scores x_i wins for each $k = 1, 2, \dots, N$. (It may happen that several adjacent x_i 's are equal. In that case fix some T_i for each.) Then the total score, say S , is simply the number of matches played. Hence

$$S = x_1 + x_2 + \dots + x_N = \binom{N}{2} = \frac{N(N-1)}{2} \quad (1)$$

For each $k = 1, 2, \dots, N$, we have to find both a lower bound and an upper bound on x_k . The case $k = 1$ is trivial. It reduces to proving that

$$\frac{N-1}{2} \leq x_1 \leq N-1 \quad (2)$$

The upper bound follows because every team plays exactly $N-1$ matches and hence also wins at most $N-1$ matches. The lower bound requires slightly more work. By (1), the average score is $\frac{N-1}{2}$. As x_1 is the maximum score, we must have $x_1 \geq \frac{N-1}{2}$. The logic here is merely that a topper must score at least as much as the average. The only way the equality can hold is when all scores are equal and hence all participants are toppers. (An example of this is a tournament among three

teams, say T_1, T_2, T_3 which beat each other cyclically, that is T_1 beats T_2 , T_2 beats T_3 and T_3 beats T_1 . Here $N = 3$ and $x_1 = x_2 = x_3 = 1$.)

Unfortunately, the case $k = 1$ is deceptively simple as it offers little clue about proving subsequent cases. Take $k = 2$. Here we have to prove that

$$\frac{N-2}{2} \leq x_2 \leq N - \frac{3}{2} \quad (3)$$

The rightmost term is not an integer, but x_2 is. So the second part actually amounts to proving that $x_2 \leq N-2$, the integral part of $N - \frac{3}{2}$. This does not follow automatically from (2) because even if $x_2 \leq x_1$, it may happen that $x_2 = x_1$. Here we proceed by contradiction. If $x_2 > N-2$, then, again by integrality of x_2 , we must have $x_2 \geq N-1$. But then x_1 and x_2 will both equal $N-1$. This means both T_1 and T_2 win every match they play. But as there is a match between T_1 and T_2 too, both cannot be winners. Hence we must have $x_2 \leq N-2$ and so $x_2 \leq N - \frac{3}{2}$ too.

The point to note is that this contradiction did not come purely algebraically, that is, from the mere knowledge that $x_1, x_2, \dots, x_N$ are integers between 0 and $N-1$ in a descending order which add to $\frac{N(N-1)}{2}$. We had to go into what x_1 and x_2 stand for, viz. the number of wins of T_1 and T_2 respectively.

Fortunately, the other part of (3), viz. getting a lower bound for x_2 does come purely algebraically, in view of $N-1$ as an upper bound for x_1 . For each $k = 1, 2, \dots, N$, let $S_k = x_k + x_{k+1} + \dots + x_N$. Then S_1 is simply S while S_2 is $S_1 - x_1$. Hence

$$S_2 = S_1 - x_1 \geq \frac{N(N-1)}{2} - (N-1) \quad (4)$$

S_2 is the sum of $N-1$ elements. The average of these $N-1$ elements is $\frac{N}{2} - 1 = \frac{N-2}{2}$. Since x_2 is the maximum among these $N-1$ elements, it is at least the average. That finishes the proof of (3).

In the next case, $k = 3$, we have to prove that

$$\frac{N-3}{2} \leq x_3 \leq N-2 \quad (5)$$

The upper bound follows because $x_3 \leq x_2 \leq N-2$ as already shown. As for the lower bound, we use the upper bounds for x_1 and x_2 already

established as $N - 1$ and $N - 2$ respectively and get a lower bound on S_3 whose largest summand is x_3 .

$$\begin{aligned}
S_3 = S_1 - (x_1 + x_2) &\geq S_1 - (2N - 3) \\
&= \frac{N(N-1)}{2} - (2N - 3) = \frac{N^2 - 5N + 6}{2} \\
&= \frac{(N-2)(N-3)}{2}
\end{aligned} \tag{6}$$

Hence the average of the $N - 2$ numbers from x_3 to x_N is at least $\frac{N-3}{2}$. Therefore the maximum element, viz. x_3 is also at least $\frac{N-3}{2}$. That completes the proof of (5).

The pattern should now be clear. We prove the upper bounds on x_k 's by some combinatorial argument and then use the upper bounds on $x_1, x_2, \dots, x_{k-1}$ to get a lower bound on S_k . Then we apply the rule that the maximum summand in S_k , viz. x_k , is at least the average of S_k .

Let us try the next case, $k = 4$ to see if this works. Here we have to prove

$$\frac{N-4}{2} \leq x_4 \leq N - \frac{5}{2} \tag{7}$$

Once again, because of the non-integrality of the rightmost term, it is equivalent to proving that $x_4 \leq N - 3$. If this is not so, then $x_4 \geq N - 2$ since x_4 is an integer. But this forces x_1, x_2 and x_3 to be at least $N - 2$. Now let $E_4 = \{T_1, T_2, T_3, T_4\}$. (We may call it 'elite 4'.) Each of these four elite teams wins $N - 2$ (or more) matches. But outside E_4 there are only $N - 4$ teams. So, at least two of these wins of each elite member must come from winning over other elite members. Summing up, we have found a set of 4 teams each of whom beats at least two teams of this set. That means there are at least 8 matches between these four players. But that is impossible because in all there can be at most $\binom{4}{2} = 6$ matches between members of a group of 4 teams.

The generalisation to any k is routine. If k is even, say $k = 2n$, then the right half of the inequality to be proved reduces to $x_{2n} \leq N - (n + 1)$ because of non-integrality of $N - \frac{2n+1}{2}$. If this fails, then each of x_1 to x_{2n} is at least $N - n$. The elite consists of the $2n$ teams from T_1 to T_{2n} .

Each of them wins at least $N - n$ matches. Outside the elite, there are only $N - 2n$ teams. Hence each of the elite members wins at least n matches against members of the elite. So there are at least $2n^2$ matches within the elite members. But that exceeds $\binom{2n}{2}$, a contradiction. The case of an odd k , say $k = 2n + 1$ follows from the case $k = 2n$ because the upper bound to be proved for x_{2n+1} is the same as the effective upper bound for x_{2n} , viz. $N - (n + 1)$.

Now let us see whether these upper bounds for $x_1, x_2, \dots, x_{k-1}$ imply the lower bound on x_k . We already did this for $k = 3$ by (6). Unfortunately, it fails for the next case, viz. $k = 4$. The upper bounds proved for x_1, x_2, x_3 are $N - 1, N - 2$ and $N - 2$ respectively. So, analogously to (6), we get

$$\begin{aligned} S_4 = S_1 - (x_1 + x_2 + x_3) &\geq S_1 - (3N - 5) \\ &= \frac{N(N-1)}{2} - 3N + 5 = \frac{N^2 - 7N + 10}{2} \end{aligned}$$

Hence the average of $x_4, x_5, \dots, x_N$ is at least $\frac{N^2 - 7N + 10}{2(N-3)}$. This does give a lower bound on x_4 , the largest summand of S_4 . But this falls slightly short of $\frac{N-3}{2}$ which is the desired lower bound on x_4 .

So, although we have proved the right half of the inequality asked, for the left half, we need a different argument. An algebraic argument fails as just shown. But we give a combinatorial argument, based again on the same common sense principle, viz. the topper is bigger than or equal to the average.

Recall that we had applied this principle to show $x_1 \geq \frac{N-1}{2}$, because T_1 was a topper in a tournament with N teams $T_1, T_2, \dots, T_N$ in which the total score was $\binom{N}{2}$. If we want to prove that $\frac{N-2}{2}$ is a lower bound for x_2 by a similar argument, we need a tournament with players $T_2, T_3, \dots, T_N$ in which T_2 is a topper. It is tempting to construct this tournament simply by removing T_1 from the original tournament (along with the matches played by T_1). The trouble is that T_2 may no longer be the topper in this reduced tournament. This will happen if in the original tournament, T_2 had beaten T_1 , because in that case the new score of T_2 will not be x_2 but $x_2 - 1$, while that of some other player who had lost to T_1 remains unaffected and hence may exceed $x_2 - 1$, if the original score of that player was also x_2 . (As illustrated above, two different players may have the same scores in a round robin tournament.)

But with a little thought, things can be salvaged even in such a scenario. Note that in the transition from the old to the new tournament, the score of no team increases. It goes down by 1 only if that team had defeated T_1 in the original tournament. So, if some team, say T_r (where $r > 2$), replaces T_2 as a topper in the new tournament, the only way this can happen is when $x_2 = x_r$ and T_2 had won over T_1 in the original tournament but T_r hadn't. Now x_r is the top score in the new tournament with $N - 1$ participants and $\binom{N-1}{2} = \frac{(N-1)(N-2)}{2}$ as the total number of matches. Hence the average score in the new tournament is $\frac{N-2}{2}$ and therefore we have that $x_r \geq \frac{N-2}{2}$. But $x_2 = x_r$ as already noted. That proves that $x_2 \geq \frac{N-2}{2}$. Of course, we already know this from (3). But that time we had used the upper bound on x_1 (viz. $x_1 \leq N - 1$), to get a lower bound on the sum S_2 . As we know, that approach fails for $k = 3$.

But our new approach works just as easily for any k . Simply remove the players $T_1, T_2, \dots, T_{k-1}$ (along with the matches they played, whether within themselves or with any of the players still retained, viz. $T_k, T_{k+1}, \dots, T_N$). There is no change in the results of the matches played between any two of these teams T_k to T_N . Let y be the highest score in this new tournament and assume that it is attained by T_r for some $r \geq k$. T_r may or may not be a topper in the original tournament. Still, since the number of matches played in the new tournament is $\binom{N-k+1}{2} = \frac{(N-k+1)(N-k)}{2}$, this is also the total score in the new tournament. As the number of participants now is $N - k + 1$, the average score is $\frac{N-k}{2}$. By the principle that the top score in any tournament is at least the average score, we have,

$$y \geq \frac{N-k}{2} \quad (8)$$

In the reduction from the original tournament to the new tournament, the score of no team can increase. If anything, it can only go down as happens when, in the original tournament, that team had beaten some of the teams from T_1 to T_{k-1} , which are now removed. The original score of T_r is x_r . So we have $x_r \geq y$. But $r \geq k$. Hence $x_k \geq x_r$. Clubbing these inequalities with (8) gives $x_k \geq \frac{N-k}{2}$. Now the assertion is proved completely. ■

An excellent problem. Although it is about inequalities, it does not

use any standard inequalities. The computations needed are elementary. What is crucially needed is common sense (about the relationship between the top score and the average score) and the reduction of an inequality in which one of the sides is an integer and the other is not. So the problem is a good test of the ability to think, especially the idea of reduction of the tournament by dropping some participants.

CONCLUDING REMARKS

After the JEE became totally objective type in 2006, one eagerly looks at the second paper of ISI B.Sc. Entrance Test for some good problems. This year, Paper 2 is a mixed bag. Q.1, Q.2 and Q.5 in Paper 2 are rather predictable as their settings are familiar. But even though the setting of Q.4 is also familiar, normally the kind of function involved in it appears in problems asking for smoothness testing. Here smoothness is given and the values at 0 are asked. Similarly, although Q.7 is a problem on volumes of solids of revolution, it allows application of common sense to reduce the computations. Q.3 is a good problem in trigonometry. Even though Q.6 and Q.8 are both on inequalities, there is no duplication of ideas.

Paper 1 also contains some good problems. The best question is Q.27 which tests not only the knowledge of the MacLaurin test for series but also the thought involved in its proof.

Number theory has a fair representation through three problems, viz. Q.12, Q.17 and Q.25. But there are three probability problems, Q.1, Q.5 and Q.15. The first is more a problem in number theory while the last one is more a problem about evaluation of limits of certain sequences. One wishes that probability had a greater representation by dropping the absolutely trivial Q.24, for example. Similarly, the relationship between Riemann sums and integrals appears in many problems. Some of these could have been replaced by probability. Also inequalities have a disproportionate representation.

Q.3 is an unusual combination of geometry and quadratic equations. Q.23 is a good problem on the binomial expansion.

As it always happens, the MCQ format, especially the one adopted by ISI where it is given that only one option is correct, has marred some otherwise good problems. A glaring example of this is Q.10 on geometry. Here, the correct answer can be identified by working out just one special case.

Then again, there are several questions in Paper 1 which properly belong to Paper 2. By putting them into Paper 1, the unscrupulous candidates get unfairly rewarded. The best problem in Paper 1, viz. Q.27 is of this type. So is Q.28, where a candidate who brings a limit inside an integral goes scot free.

The total absence of matrices and determinants is shocking. If this is because these topics are not in the syllabus, it is high time to change the syllabus.